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Fair Skill Brier Score: Evaluating Probabilistic Forecasts of One-Off Events with Different Numbers of Categorical Outcomes

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Experts' abilities to make accurate probabilistic forecasts are often evaluated with proper scoring rules. In practice, the Brier score is one of the most commonly used scoring rules when the target outcomes are categorical. However, given that an event with more outcomes is typically more diffcult to forecast, it is unfair to compare the Brier scores of experts who forecast events with different numbers of outcomes. In this paper, we introduce a simple fair skill adjustment to the Brier score to refne such comparisons. To demonstrate our adjustment, we introduce a behavioral model of experts' probabilistic forecasts of one-off events with different numbers of categorical outcomes. Under this model, we show that the fair skill Brier score is a more reliable measure of experts' forecasting skills as long as the outcomes are not close to being certain. Finally, we apply the fair skill Brier score to experimental data from two different studies by the US intelligence community and find empirical support for our theoretical results.

Keywords: Brier Score; Forecast Verification; Probabilistic Forecast; Proper Scoring Rule; Skill Score

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Experts' abilities to make accurate probabilistic forecasts are often evaluated with proper scoring rules. In practice, the Brier score is one of the most commonly used scoring rules when the target outcomes are categorical. However, given that an event with more outcomes is typically more difficult to forecast, it is unfair to compare the Brier scores of experts who forecast events with different numbers of outcomes. In this paper, we introduce a simple fair skill adjustment to the Brier score to refine such comparisons. To demonstrate our adjustment, we introduce a behavioral model of experts' probabilistic forecasts of one-off events with different numbers of categorical outcomes. Under this model, we show that the fair skill Brier score is a more reliable measure of experts' forecasting skills as long as the outcomes are not close to being certain. Finally, we apply the fair skill Brier score to experimental data from two different studies by the US intelligence community and find empirical support for our theoretical results.

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1. Introduction

The world is filled with uncertain events, including pandemics, global conflicts, and economic instabilities. Accurate forecasts of unprecedented, high-impact events are critical for governments to make good decisions (Tetlock and Gardner 2016). The form of the target events largely determines the format of these forecasts. For events with categorical outcomes, forecasts are usually either categorical (choosing the most likely outcome) or probabilistic (estimating the likelihood of each potential outcome). For example, a pre-election forecast of a political campaign might predict the most favored candidate (a categorical forecast) or estimate each candidate's chance of winning (a probabilistic forecast). Unlike categorical forecasts, probabilistic forecasts reflect the experts' level of confidence in each one of the outcomes (Sanders 1963, Savage 1971). Such forecasts reveal more information about the experts' forecasting skills and hence can help decision-makers to identify skilled experts (e.g., Murphy and Winkler 1992, Winkler 1994), aggregate their opinions (e.g., Ranjan and Gneiting 2010, Satopää 2022), and train them to become even better forecasters (e.g., Mellers et al. 2014, Mellers et al. 2015).

The literature has proposed many scoring rules to evaluate the quality of probabilistic forecasts. The usual recommendation is to use *strictly proper* scoring rules that elicit experts' true beliefs (Gneiting and Raftery 2007, Bröcker and Smith 2007). The most popular strictly proper scoring rule for probabilistic forecasts is the Brier score, which is simply the squared distance between the probabilistic forecast and the event indicator (Brier 1950, Winkler 1994). Its popularity is likely to stem from its interpretability, simplicity, and boundedness.[1]

The expected Brier score is determined by the expert's skill to make accurate forecasts and the difficulty of forecasting the event. Even though there can be many factors contributing to the difficulty of the task, the literature often mentions the base rate of the event as an important determinant of difficulty (Murphy and Winkler 1987, Murphy and Winkler 1992). For instance, an expert is more likely to achieve a smaller Brier score in forecasting a binary event with a base rate of 1% than one with a base rate of 50%. This makes it difficult to compare experts' scores in the common context where they forecast different events with very different base rates (Winkler 1994). If the base rates are known, it becomes possible to normalize varying difficulty of the events by comparing the expert's score against the score that one would get by always forecasting the base rate (e.g., Murphy 1973, Murphy 1974, Winkler 1994). Unfortunately, judgment elicitation tasks typically focus on one-off events for which the base rates are unknown (O'Hagan et al. 2006, Chapter 3).

In this paper, our emphasis shifts from unequal base rates to other more tangible factors that contribute to varying difficulty within the one-off setting. In particular, we consider experts who forecast events with different numbers of outcomes. Practical situations like these are quite prevalent. For instance, crowd forecasting platforms, such as Metaculus[2] and Good Judgment Open[3], feature events with varying numbers of possible outcomes. Such variation can make experts' scores of forecasting different events incomparable because the complexity and hence the difficulty of forecasting an event often changes with the number of outcomes. For example, it is often harder to forecast an election with more eligible candidates because the votes would be split among more competitors.

[1] Unlike the logarithmic score (Good 1952), the Brier score is always bounded, which allows experts' average Brier scores to recover from a single large forecast error.

[2] For more information, see `https://www.metaculus.com/questions/`.

[3] For more information, see `https://www.gjopen.com/questions`.

To compare experts who forecast events with different numbers of outcomes, we propose the *fair skill Brier score* which is simply the difference between the expert's Brier score and the Brier score that the expert would get by always forecasting each outcome to be equally likely.[4] This adjusted score has several attractive properties. First, given that the adjustment does not depend on experts' forecasts, our fair skill Brier score remains strictly proper. Second, given that the Brier score of a uniform forecast increases with the number of outcomes, an expert who forecasts an event with more potential outcomes gets compensated for the added difficulty of the task. We argue that this leads to a more accurate and hence fairer[5] comparison of the experts' forecasting skills.[6]

Past research compares scoring rules either axiomatically (e.g., Selten 1998, Jose 2009, Machete 2013) or empirically (e.g., Bickel 2007, Merkle and Steyvers 2013). To the best of our knowledge, the current paper is the first to make a theoretical argument based on statistical data analysis that seeks to estimate a population mean based on a finite sample. In judgmental forecasting, experts are scored based on their forecasts of a small number of events. These *sample scores* are then used as proxies for the experts' forecasting skills that, in this paper, represent their long-term (or expected) performance. Unfortunately, the correlation between experts' sample scores and their long-term performance is likely to depend on the decision-maker's choice of the exact scoring rule. In the current work, we show that the fair skill adjustment to the Brier score improves this correlation metric if, in general, the experts' expected Brier scores increase with the number of outcomes at least half as fast as the Brier score of the uniform forecast.

To gain a better understanding of exactly when and by how much the fair skill adjustment can be expected to improve our correlation metric, we introduce a behavioral model of experts' probabilistic forecasts of categorical outcomes. This model allows us to explore the benefits of fair skill adjustment in terms of the heterogeneity of the number of outcomes of the events, the level of the inherent uncertainty in the events, the level of information diversity in the crowd, the level of calibration in the experts' information processing, and the number of forecasts included

[4] A similar adjustment to the logarithmic score was briefly mentioned by Benedetti (2010). Their motivation, however, was to suggest a default base rate for skill scores – not to study or control for heterogeneity of difficulty that arises from different numbers of outcomes.

[5] Our hope is to avoid philosophical debates regarding the definition of fairness. In this context, we use the term "fair skill" to refer to a score that is more likely to reflect the true skill of the experts. Following the term "fair skill score" in Benedetti (2010), we name our score fair skill Brier score.

[6] Other baselines except the uniform forecast have been studied in the past. For instance, Lambert et al. (2008) used the average score of a fixed group of experts as their baseline. However, in a more general context where experts may forecast different sets of events, it is challenging to identify any baseline other than the uniform forecast that is universally applicable, interpretable, and fixed ex-ante. When experts forecast different (potentially overlapping) sets of events, using the average score as a baseline is likely to raise fairness concerns because the "relative" skill score now depends on the group of experts forecasting each question. For instance, an expert with average skill may appear stellar if they happen to make forecasts of the same events as a group of poorly skilled experts. For a numerical example illustrating the fairness concerns in using the average score as the baseline, see Appendix A.

in the calculation of the sample Brier score. Considering a wide range of practically plausible environments, we find that the fair skill adjustment improves our correlation metric as long as the events are not close to being certain (in which case one may argue that consulting experts is hardly necessary anyway). The improvement is more significant when the experts are either under- or over-confident and increases as the numbers of outcomes of the events become more heterogeneous or as the sample of past scores grows from a small to a medium size (e.g., from 10 to 50).

To offer empirical support for our fair skill adjustment, we apply this method to experimental data from two large-scale studies conducted by Intelligence Research Advanced Projects Activity (IARPA). These studies collectively involve thousands of experts making probabilistic forecasts for hundreds of events with categorical outcomes. In the first study, experts were asked to make probabilistic forecasts of simulated events with different numbers of outcomes. In the second study, experts were asked to make probabilistic forecasts of real-life geopolitical events with different numbers of outcomes. Overall, we see improvements in our correlation metric in both studies, and the empirical findings are consistent with our theoretical results.

The rest of this paper is organized as follows. Section 2 introduces the fair skill Brier score and explains intuitively why it can help to control for the heterogeneity of difficulty due to different numbers of outcomes. Section 3 introduces our behavioral model of experts' forecasts and explores when and by how much the fair skill adjustment improves the correlation coefficient between the experts' sample and long-term scores. Section 4 applies the fair skill Brier score to empirical data and finds support for our theoretical results. Finally, Section 5 concludes the paper and discusses future research. Appendix holds all proofs, longer derivations, additional illustrations, and robustness checks of our main results.

2. Fair Skill Brier Score

2.1. Aligning Expected Scores

Consider an event with B categorical outcomes. Suppose an expert reports probabilistic forecasts $f_1, \ldots, f_B$ for these outcomes. If the b^*th outcome happens, then the expert's Brier score is

$$\mathrm{BrS} = \sum_{b=1}^{B} (f_b - y_b)^2 = 1 - 2f_{b^*} + \sum_{b=1}^{B} f_b^2, \tag{1}$$

where $y_b = 1$ for $b = b^*$ is the materialized outcome and $y_b = 0$ for $b \neq b^*$. The score ranges between 0 and 2 and is negatively orientated. In particular, an omniscient expert would report $f_b = y_b$ for $b = 1, \ldots, B$ and receive a Brier score of 0, whereas an expert who assigns 100% probability to an outcome that does not materialize receives the maximum Brier score of 2.

Different experts' average Brier scores can be directly compared if the experts forecast the same set of events. However, comparisons may become unfair if they forecast different events and some

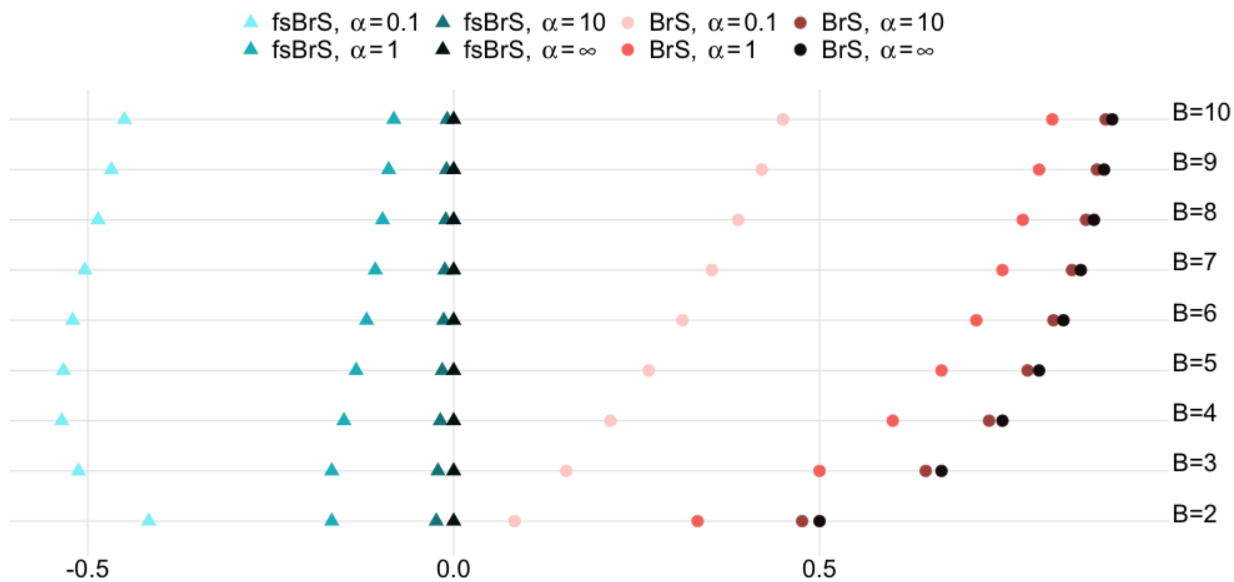

Figure 1 **The circles indicate the perfectly skilled expert's average Brier scores (BrS) for different numbers of outcomes B and different values of the concentration parameter α that controls the distribution of the underlying probabilities of the event outcomes (see Figure 2) and hence the level of the inherent uncertainty in the events. The triangles represent the perfectly skilled expert's average fair skill Brier scores (fsBrS). The triangles show overall better vertical alignment than the circles.**

experts face more challenging forecasting tasks than others. We denote the underlying probabilities of the B outcomes of an event by $\boldsymbol{p}_B = (p_1, \ldots, p_B)$. Given that the Brier score is strictly proper, its expectation is minimized by underlying probabilities. Then, conditional on B, the expected Brier score of a perfectly skilled expert who forecasts the underlying probabilities $f_b = p_b$ for all $b = 1, \ldots, B$ is

$$\mathbb{E}\left[\mathrm{BrS}_{\text{perfect-skill}} | \boldsymbol{p}_B, B\right] = 1 - 2\sum_{b=1}^{B} f_b p_b + \sum_{b=1}^{B} f_b^2 \bigg|_{f_b = p_b} = 1 - \sum_{b=1}^{B} p_b^2. \tag{2}$$

The vector of underlying probabilities $\boldsymbol{p}_B$ may vary across different events. To illustrate, suppose $\boldsymbol{p}_B = (p_1, \ldots, p_B)$ is randomly drawn from a symmetric Dirichlet distribution with concentration parameter α. This parameter controls the concentration of the Dirichlet distribution of $\boldsymbol{p}_B$ around the uniform probabilities $(1/B, \ldots, 1/B)$. Figure 2 shows this effect for probabilities of events with three outcomes. After we integrate out $\boldsymbol{p}_B$ from (2), the conditional Brier score depends only on the number of outcomes B and the concentration parameter α. Figure 1 considers this Brier score under different values of α and shows how it increases with the number of potential outcomes B. Therefore, even a perfectly-skilled expert may find it unfair to have their Brier score compared directly against the Brier score of another expert who forecasted events with fewer outcomes.

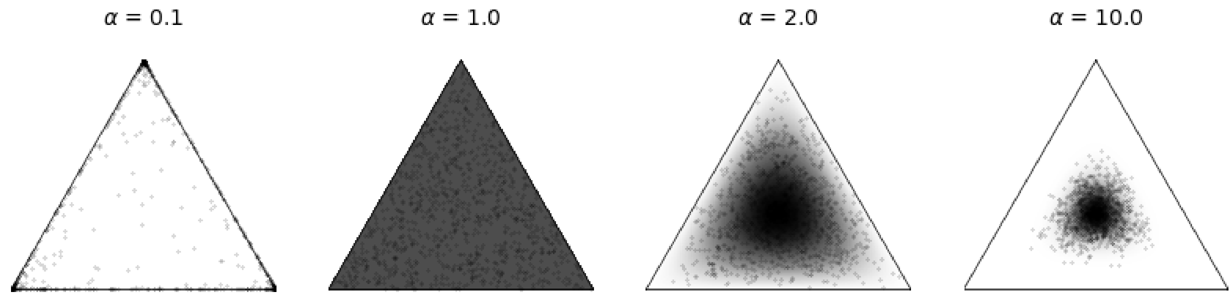

Figure 2 **The parameter α describes how concentrated the distribution of $\boldsymbol{p}_B$ is around the uniform probability $(1/B, \ldots, 1/B)$ at the center. In particular, when $\alpha = 1$, the probability $\boldsymbol{p}_B$ is uniformly distributed over the simplex. But when $\alpha > 1$ or $\alpha < 1$, the probability $\boldsymbol{p}_B$ is more or less, respectively, concentrated around the center.**

To make such scores more comparable, we introduce the "fair skill" adjustment. This compensates experts for forecasting events with more potential outcomes. The level of compensation increases monotonically with the number of potential outcomes.

Definition 1 (Fair Skill Brier Score). Consider an event with B categorical outcomes. Suppose the expert's Brier score for this event is BrS, then the corresponding fair skill Brier score is

$$\mathrm{fsBrS} = \mathrm{BrS} - (B-1)/B. \tag{3}$$

The adjustment $(B-1)/B$ in (3) is the expected score of an unskilled expert who is guided by the principle of insufficient reason (or the "principle of indifference") and always forecasts all B outcomes to be equally likely (Keynes 1921), i.e., $f_b = 1/B$ for all $b = 1, \ldots, B$. The fair skill Brier score then represents the added forecasting accuracy compared to an unskilled expert. Moreover, given that the adjustment does not depend on the experts' forecasts or any unknown quantities, it is simple to use and easy to explain to experts. Furthermore, fair skill adjustment does not change the experts' behavior. Therefore, as shown by Proposition 1, the fair skill Brier score remains strictly proper.

Proposition 1. *The fair skill Brier score is strictly proper.*

In Figure 1 the adjustment terms at each value of B between 2 and 10 are given by the right-most circles (corresponding to BrS at $\alpha = \infty$). The triangles on the left side of the figure then show the mean values of the resulting fair skill Brier scores at different values of B and α. These are vertically more aligned than the circles, suggesting that, at least in this illustration, fair skill adjustment can be an effective mechanism for countering the effects of increasing numbers of outcomes.

2.2. Short-Term and Long-Term Scores

Consider multiple experts indexed by $k = 1, 2, \ldots$. Ideally, we would rank the experts based on their true forecasting skills. A typical approach to achieving this goal is to ask each expert to forecast, e.g., N events and then rank them based on their average scores. A low-skill expert, however, can end up performing relatively well if they happen to guess the correct answers by chance. Such lucky guessing becomes more difficult as the number of events, N, grows larger. In fact, the role of chance would be removed by ranking the experts based on their long-term performances.

To make this more specific, consider some generic scoring rule S and suppose that expert k's scores $S_{k,1}, S_{k,2}, \ldots$ for forecasting different events are drawn independently from some distribution. The best estimator of expert k's future performance then is the long-term or expected score $\mathbb{E}[S_k]$. However, given that the long-term scores are unknown in practice, it becomes important to understand how closely the skill ranking can be reconstructed based on the available sample scores. This, in turn, depends on how strongly these sample scores correlate with the long-term scores. To inspect this further, denote expert k's sample score with $\overline{S}_k = \frac{1}{N}\sum_{n=1}^{N} S_{k,n}$. The Pearson correlation[7] between the experts' sample scores $\overline{S}$ and long-term scores $\mathbb{E}[S]$ is

$$\mathrm{Corr}\left(\overline{S}, \mathbb{E}[S]\right) = \frac{\mathrm{Cov}\left(\overline{S}, \mathbb{E}[S]\right)}{\sqrt{\mathrm{Var}\left(\overline{S}\right)\mathrm{Var}\left(\mathbb{E}[S]\right)}}. \tag{4}$$

Given that $\mathbb{E}[\mathrm{fsBrS}]$ differs from $\mathbb{E}[\mathrm{BrS}]$ only by a constant, namely by the expected adjustment factor across the entire population of events, we have that

$$\mathrm{Corr}\left(\overline{S}, \mathbb{E}[\mathrm{fsBrS}]\right) = \mathrm{Corr}\left(\overline{S}, \mathbb{E}[\mathrm{BrS}]\right) \tag{5}$$

for both $\overline{S} = \overline{\mathrm{BrS}}$ and $\overline{S} = \overline{\mathrm{fsBrS}}$. In other words, in our correlation analysis, it does not matter whether we consider the long-term score to be the expected Brier score or the expected fair skill Brier score. For simplicity and because the Brier score is the typical score used in the literature, we will denote the long-term score with the expected Brier score.

Our goal is then to understand whether $\mathrm{Corr}\left(\overline{\mathrm{fsBrS}}, \mathbb{E}[\mathrm{BrS}]\right)$ is greater than $\mathrm{Corr}\left(\overline{\mathrm{BrS}}, \mathbb{E}[\mathrm{BrS}]\right)$. As (4) shows, this depends on (i) how much the sample scores co-vary with the long-term scores and (ii) how variable these sample scores are around their own mean. First, consider condition (i). Denote expert k's sample fair skill Brier with $\overline{\mathrm{fsBrS}}_k = \overline{\mathrm{BrS}}_k - \overline{\mathrm{U}}_k$, where $\overline{\mathrm{U}}_k$ is the average adjustment of the N individual Brier scores in the sample. Given that $\mathrm{Cov}\left(\overline{\mathrm{fsBrS}}, \mathbb{E}[\mathrm{BrS}]\right) =$

[7] Given that the Pearson correlation is more tractable in our upcoming analysis, we will focus now on the Pearson correlation and refer to the Spearman correlation in robustness checks. In Appendix D, we show that the results under the Spearman correlation resemble those under the Pearson correlation, indicating that the fair skill adjustment improves the decision-maker's ability to identify skilled experts, even if they are only interested in the ranking.

$\text{Cov}\left(\overline{\text{BrS}}, \mathbb{E}\left[\text{BrS}\right]\right) - \text{Cov}\left(\overline{\text{U}}, \mathbb{E}\left[\text{BrS}\right]\right)$, we have that $\text{Cov}\left(\overline{\text{BrS}}, \mathbb{E}\left[\text{BrS}\right]\right) \leqslant \text{Cov}\left(\overline{\text{fsBrS}}, \mathbb{E}\left[\text{BrS}\right]\right)$ if and only if

$$\text{Cov}\left(\overline{\text{U}}, \mathbb{E}\left[\text{BrS}\right]\right) \leqslant 0. \tag{6}$$

Condition (6) holds as long as less skilled individuals do not typically forecast events with more outcomes. This is likely to hold when the experts' true forecasting skills are not known beforehand or the experts are assigned to events at random.

Next, consider condition (ii). If (6) holds, then $\text{Corr}\left(\overline{\text{fsBrS}}, \mathbb{E}\left[\text{BrS}\right]\right) > \text{Corr}\left(\overline{\text{BrS}}, \mathbb{E}\left[\text{BrS}\right]\right)$ if $\text{Var}\left(\overline{\text{fsBrS}}\right) < \text{Var}\left(\overline{\text{BrS}}\right)$. By the law of total variance, we have that $\text{Var}\left(\overline{S}\right) = \mathbb{E}\left[\text{Var}\left(\overline{S}_k\right)\right] + \text{Var}\left(\mathbb{E}\left[\overline{S}_k\right]\right)$. Given that $\text{Var}\left(\mathbb{E}\left[\overline{\text{fsBrS}}_k\right]\right) = \text{Var}\left(\mathbb{E}\left[\overline{\text{BrS}}_k - \overline{\text{U}}_k\right]\right) = \text{Var}\left(\mathbb{E}\left[\overline{\text{BrS}}_k\right]\right)$, we have that $\text{Var}\left(\overline{\text{fsBrS}}\right) < \text{Var}\left(\overline{\text{BrS}}\right)$ if and only if $\mathbb{E}\left[\text{Var}\left(\overline{\text{fsBrS}}_k\right)\right] < \mathbb{E}\left[\text{Var}\left(\overline{\text{BrS}}_k\right)\right]$, that is, if and only if, in expectation, an expert's sample fair skill Brier score is less variable than their sample Brier score. Given that the N scores in the sample are independent and identically distributed, it is sufficient to check whether $\text{Var}\left(\text{fsBrS}_k\right) < \text{Var}\left(\text{BrS}_k\right)$ holds in expectation. Again, using the law of total variance, we can decompose this variance of an individual's score by conditioning on the number of outcomes B of the event:

$$\text{Var}\left(S_k\right) = \underbrace{\mathbb{E}\left[\text{Var}\left(S_k|B\right)\right]}_{\text{Within-Component Variation}} + \underbrace{\text{Var}\left(\mathbb{E}\left[S_k|B\right]\right)}_{\text{Cross-Component Variation}}.$$

Given that $\text{Var}\left(\text{fsBrS}_k|B\right) = \text{Var}\left(\left(\text{BrS}_k - \text{U}_k\right)|B\right) = \text{Var}\left(\text{BrS}_k|B\right)$ for fixed B, we have that $\text{Var}\left(\text{fsBrS}_k\right) < \text{Var}\left(\text{BrS}_k\right)$ if and only if

$$\text{Var}\left(\mathbb{E}\left[\text{fsBrS}_k|B\right]\right) < \text{Var}\left(\mathbb{E}\left[\text{BrS}_k|B\right]\right). \tag{7}$$

Along with our discussion in Section 2.1, the efficiency of our adjustment then depends on how well it can align the expected scores at different numbers of outcomes B. As long as our adjustment improves the alignment on average across the population of experts, we expect our fair skill Brier score to be a better measure of the experts' long-term performance. Clearly, for this to work, the adjustment should increase with B similarly to the way that the expert's expected Brier score increases with B. The following proposition returns to condition (7) and shows that the meaning of "similarly" can be interpreted here rather loosely.

PROPOSITION 2. $\text{Var}\left(\mathbb{E}\left[\text{fsBrS}_k|B\right]\right) < \text{Var}\left(\mathbb{E}\left[\text{BrS}_k|B\right]\right)$ *is true for all distributions of the number of outcomes B if and only if the expert's (expected) Brier score increases with B at least half as fast as the Brier score of the uniform forecast, that is, if for all B we have that* $\mathbb{E}\left[\text{BrS}_k|\left(B+1\right)\right] - \mathbb{E}\left[\text{BrS}_k|B\right] > 0.5\left[B/\left(B+1\right) - \left(B-1\right)/B\right]$.

To illustrate, in Figure 1 the triangles representing the expected fair skill Brier scores show better vertical alignment than the circles representing the expected Brier scores for all B when $\alpha = 1, 10$, or ∞. This is when the perfectly skilled expert's expected Brier score increases fast enough with B. However, suppose the distribution of the underlying probabilities is heavily concentrated at one of the outcomes (e.g., when $\alpha = 0.1$). In that case, the perfectly skilled expert is nearly omniscient, their expected Brier scores are close to 0 for all B and hence will not increase fast enough with B to benefit from our fair skill adjustment. Nevertheless, this considers an ideal expert under extreme conditions. In practice, experts are often only partially skilled. Their scores are then likely to be more similar to the scores of the uniform forecast, suggesting a tendency towards conditions where the fair skill adjustment is likely to improve our inference of experts' true forecasting skills.

3. Behavioral Model of Experts' Forecasts

3.1. Preliminaries

To better understand when and by how much the fair skill adjustment can be expected to improve our ability to capture the experts' long-term performance, we develop a behavioral model of experts making probabilistic forecasts of events with different numbers of categorical outcomes. Our model is tractable and extends the popular "measurement error models" (e.g., Morris and Shin 2002, Gaba et al. 2018, Palley and Satopää 2023) to experts who forecast different sets of events with different numbers of outcomes. The model is parameterized in terms of critical features of the forecasting environment, including the level of the inherent difficulty of the forecasting tasks, information diversity and the level of calibration among experts, number of sample scores, and heterogeneity of the number of outcomes of the events. By varying the parameters, we can then explore how the advantage of our fair skill adjustment depends on the context.

3.2. Events

To model events, we assume that the vector of underlying probabilities $\boldsymbol{p}_B = (p_1, \ldots, p_B)$ follows the Dirichlet distribution[8] and that the number of outcomes B follows some distribution π_B that is, for now, kept general. An outcome $\boldsymbol{y}_B = (y_1, \ldots, y_B)$ is then generated as follows:

$$\boldsymbol{y}_B | \boldsymbol{p}_B \sim \text{Categorical}(\boldsymbol{p}_B), \tag{8}$$

$$\boldsymbol{p}_B | \boldsymbol{\alpha}_B \sim \text{Dirichlet}(\boldsymbol{\alpha}_B), \tag{9}$$

$$B \sim \pi_B, \tag{10}$$

[8] Different draws of $\boldsymbol{p}_B$ are assumed to be uncorrelated. This is because asking individuals to forecast dependent events reduces the effective sample size of questions. To draw an analogy, consider evaluating students' understanding of a course. To assess their overall comprehension, it is preferable to use independent questions rather than ones where the correct answer to one question provides a clue to the correct answer to another. Therefore, to capture experts' true skills efficiently, correlated events should be avoided in practice. This has been discussed by Satopää et al. (2021).

where $\boldsymbol{\alpha}_B = (\alpha_1, \dots, \alpha_B) \in \mathbb{R}_+^B$ is the parameter vector of the Dirichlet distribution. If any α_b is larger than the other α_b's, the corresponding p_b is more likely to be higher than the other p_b's. Given that the potential outcomes of the event are not ordered, the labeling of $y_1, \dots, y_B$ and hence of $p_1, \dots, p_B$ is arbitrary. We can then organize (e.g., by permuting the outcome labels of each event uniformly at random) the orders of the potential outcomes of each event in the population such that the distribution of the underlying probabilities becomes symmetric around the uniform probabilities. Under our Dirichlet distribution, this implies $\alpha_1 = \dots = \alpha_B = \alpha$. The *concentration* parameter α describes how concentrated the underlying probabilities are around the uniform distribution and hence controls the level of the inherent uncertainty in the events. This is illustrated in Figure 2 for three-outcome events.

3.3. Perfectly Skilled Expert

In this section, we return to our discussion of the perfectly skilled expert in Figure 1. To begin, the following proposition computes the expected Brier score of a perfectly skilled expert.

PROPOSITION 3. *If the underlying probabilities follow a symmetric Dirichlet distribution with concentration parameter α, then, conditional on the number of outcomes B, the expected Brier score of a perfectly skilled expert is*

$$\mathbb{E}\left[\mathrm{BrS}_{\textit{perfect-skill}} | \alpha, B\right] = 1 - (1+\alpha) / (1 + B\alpha). \tag{11}$$

The expected Brier score (11) increases with the number of potential outcomes B and with the concentration parameter α. On one hand, as the level of the inherent uncertainty increases ($\alpha \to \infty$), the expected Brier score approaches the fair skill adjustment term, resulting in better alignment and hence larger benefits of our adjustment. On the other hand, as the level of the inherent uncertainty vanishes ($\alpha \to 0$), the perfectly skilled expert becomes omniscient and hence receives the perfect Brier score of 0 at all values of B (recall Figure 1). The expected Brier scores conditional on different values of B are then perfectly aligned around 0, and fair skill adjustment only introduces extra variability.

Admittedly, the extreme value $\alpha = 0$ is hardly realistic as it assumes no inherent uncertainty among the different outcomes of an event. A natural question then is: What is the smallest value of α at which the fair skill adjustment can still be expected to reduce variability in the conditional scores of a perfectly skilled expert? A precise answer is given in the next proposition.

PROPOSITION 4. *Under our symmetric Dirichlet model,* $\mathrm{Var}\left(\mathbb{E}\left[\mathrm{fsBrS}_{\textit{perfect-skill}} | B\right]\right) < \mathrm{Var}\left(\mathbb{E}\left[\mathrm{BrS}_{\textit{perfect-skill}} | B\right]\right)$ *if* $\alpha > \left(\sqrt{73} - 7\right)/12 \approx 0.13$.

The cutoff value $\alpha \approx 0.13$ represents scenarios where there is very little uncertainty among the outcomes of an event (recall Figure 2). In practice, experts are often asked to forecast events with substantial uncertainty. For instance, in our second empirical study, experts were asked to make probabilistic forecasts of real-world events chosen by the intelligence community. A crucial selection criterion for the events was that, before posting, each outcome should be deemed more than 10% but less than 90% likely.[9] Given this rule, our Dirichlet model with the cutoff value for α given in Proposition 4 would generate acceptable binary events only around 23.1% of the time. This illustrates how our cutoff value is unlikely to represent the events that are the typical focus in judgmental forecasting exercises.

3.4. Partially Skilled Experts

Experts often lack calibration and make forecasts with partial information (Satopää et al. 2021). Different degrees of calibration and levels of information then generate diverse forecasting skills among the experts. To bring such heterogeneity into our Dirichlet model, suppose that the expert makes a forecast after observing I independent draws from the true distribution of the potential outcomes (8). Similarly to the number of potential outcomes B in Section 3.2, we assume that the expert's amount of information I is drawn from some distribution π_I. To capture different degrees of calibration, suppose the expert's prior distribution for the underlying probabilities $\boldsymbol{p}_B$ is a Dirichlet distribution with $\boldsymbol{\beta}_B = (\beta, \dots, \beta)$, where β may not equal to α.

If x_b represents the number of times that the expert observed the bth outcome, the total number of observed outcomes $\boldsymbol{x}_B = (x_1, \dots, x_B)$ is a multinomial random variable with I trials each with probabilities $\boldsymbol{p}_B$. Given that the Dirichlet distribution is the conjugate prior for the multinomial distribution and the expert's prior distribution of $\boldsymbol{p}_B$ is Dirichlet with a parameter vector $\boldsymbol{\beta}_B$, the expert's posterior distribution of $\boldsymbol{p}_B$ is a Dirichlet distribution with an updated parameter vector $\boldsymbol{\beta}_B + \boldsymbol{x}_B$. To minimize a proper scoring rule, the expert reports the posterior expectation of $\boldsymbol{p}_B$ as their forecast.

In summary, conditional on the number of potential outcomes B and the underlying probabilities $\boldsymbol{p}_B$, the expert's forecast is generated as follows:

$$\boldsymbol{f}_B = \mathbb{E}\left[\boldsymbol{p}_B | \boldsymbol{x}_B\right] = \frac{\boldsymbol{\beta}_B + \boldsymbol{x}_B}{B\beta + I}, \tag{12}$$

$$\boldsymbol{x}_B | I, \boldsymbol{p}_B \sim \text{Multinomial}\left(I, \boldsymbol{p}_B\right), \tag{13}$$

$$I \sim \pi_I. \tag{14}$$

Unlike the perfectly skilled expert whose expected Brier score only depends on the event characteristics α and B (recall Proposition 3), the partially skilled expert's expected Brier score is also

[9] Program website: https://www.iarpa.gov/research-programs/ace.

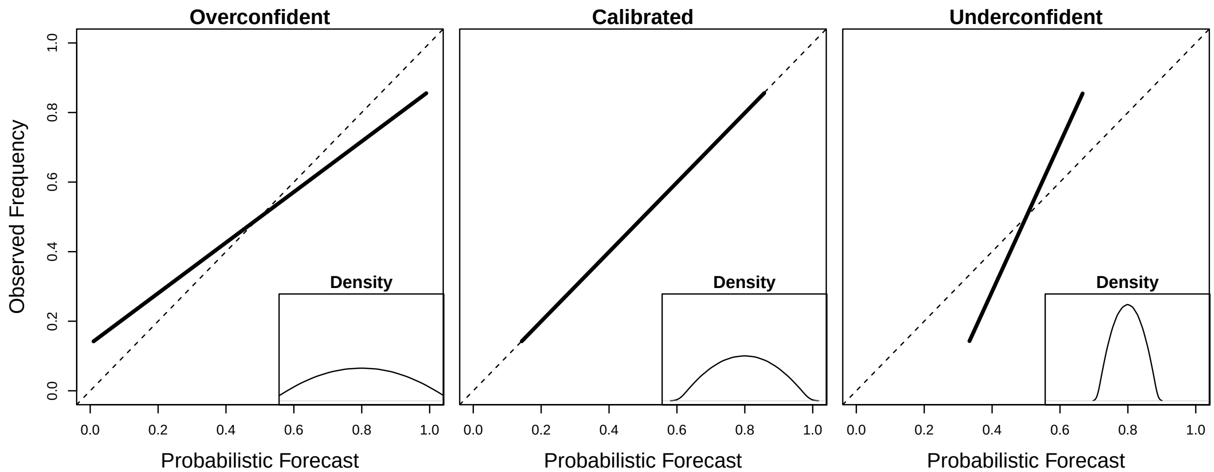

Figure 3 **Calibration plots of experts' probabilistic forecasts of events with $B=2$ outcomes under increasing values of β ($\beta=0.1,2,10$ from left to right). The information level is $I=10$. We fix $\alpha=2$ so that in the left panel, the expert is over-confident ($\beta<\alpha$), in the middle, the expert is calibrated ($\beta=\alpha$), and in the right panel, the expert is under-confident ($\beta>\alpha$). The inlaid density plots show the distributions of experts' probabilistic forecasts.**

affected by their skill level as described by the expert characteristic I and β. The next proposition makes this specific.

PROPOSITION 5. *Conditional on α, β, B, and I, the expert's expected Brier score is*

$$\mathbb{E}\left[\mathrm{BrS}_k|\alpha,\beta,B,I\right]=\frac{B-1}{B\alpha+1}\left(\frac{B\beta^2+I\alpha}{\left(B\beta+I\right)^2}+\alpha\right). \tag{15}$$

Assuming that $I>0$, the expected score (15) is lower than $(B-1)/B$ if and only if $\alpha<2\beta+I/B$.

The expert's expected Brier score (15) generally decreases as they acquire more information, i.e., as I increases. When $I\to\infty$, the effect of a poorly chosen prior (i.e., β far from α) is "washed away", and the expected Brier score coincides with that of a perfectly skilled expert (11). Of course, holding all else fixed, the expected Brier score (15) is minimized when the expert's prior distribution aligns with the actual outcome-generating process (i.e., when $\beta=\alpha$). However, if $\beta>\alpha$ (or $\beta<\alpha$), then the expert places too much (little) weight on the prior belief and is hence under-confident (over-confident) in the acquired information. Figure 3 illustrates these effects with calibration plots under $I=10$, $\alpha=2$, and three different levels of β, namely $\beta=0.1$ on the left, $\beta=2$ in the middle, and $\beta=10$ on the right. The inlaid density plots show the distributions of the corresponding forecasts.

The second part of Proposition 5 shows that the experts outperform, in expectation, the uniform forecast if they are not highly over-confident ($\beta>\alpha/2$) or they have sufficient amounts of information per outcome to overcome their over-confidence and the inherent uncertainties of the multiple competing outcomes ($I/B > \alpha - 2\beta$).

Given that our behavioral model represents a wide range of different forecasting environments, it allows us to study when and by how much the fair skill adjustment can be expected to improve the correlation between the experts' sample scores and long-term scores:

$$\Delta\mathrm{Corr} = \mathrm{Corr}\left(\overline{\mathrm{fsBrS}}, \mathbb{E}\left[\mathrm{BrS}\right]\right) - \mathrm{Corr}\left(\overline{\mathrm{BrS}}, \mathbb{E}\left[\mathrm{BrS}\right]\right).$$

Even though it is possible to derive a closed-form expression for ΔCorr under our Dirichlet model, this derivation is complicated and hence has been deferred to Appendix B.6. Instead, in this section, we will visualize ΔCorr over a range of forecasting environments. First, we make specific choices for the distributions of the number of outcomes and experts' information in (10) and (14), respectively. In terms of information, we consider experts whose level of information is drawn from the uniform distribution between some given values $I_{\min}$ and $I_{\max}$. Similarly, each expert forecasts an independent set of N events for which the numbers of potential outcomes are drawn from the uniform distribution between some given values $B_{\min}$ and $B_{\max}$.

Figure 4 fixes the minimum level of information to $I_{\min} = 10$, the number of sample scores to $N = 10$, and the minimum number of potential outcomes to $B_{\min} = 2$. Under these choices, the figure then illustrates how ΔCorr changes with α, β, $I_{\max}$, and $B_{\max}$. In particular, the concentration parameter α increases from 0.1 in the left-most column to 1.0 and 2.0 in the middle and right-most columns, respectively. The experts' prior parameter β increases from 0.1 in the top-most row to 1.0 and 2.0 in the middle and bottom-most rows, respectively. Therefore, the inherent uncertainty of the events increases panel-by-panel from left to right, and the experts' confidence in their information decreases from top to bottom. Within each panel, $B_{\max}$ changes on the x-axis and $I_{\max}$ changes on the y-axis. This means that the events become more diverse regarding the number of outcomes towards the right of each panel, and the information diversity among the experts increases towards the top of each panel.

Figure 4 shows that fair skill adjustment is more helpful when experts lack calibration (i.e., $\beta \neq \alpha$). Intuitively, this makes sense because poorly calibrated experts' Brier scores are likely to resemble and hence align more closely with those of a uniform forecaster. When experts happen to be calibrated (i.e., $\beta = \alpha$), the improvement in the correlation metric increases with the level of the inherent uncertainty of the events. This is consistent with what we have found in Proposition 4. Indeed, elicitation exercises in practice often focus on events in the "Goldilocks zone" of difficulty (recall our discussion at the end of Section 3.3), where they are neither trivial nor impossible to forecast. Our results suggest that it is in such environments that fair skill adjustment can

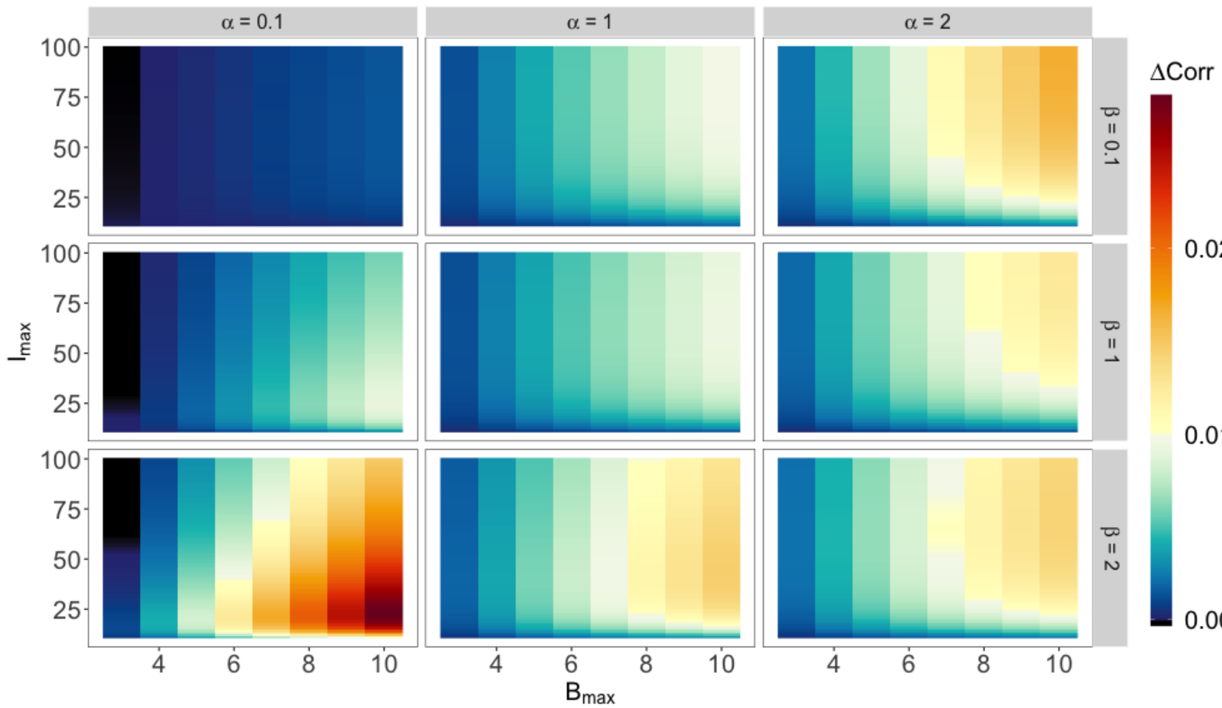

Figure 4 **The amounts by which fair skill adjustment improves the Pearson correlation between the experts' sample ($N = 10$) and long-term scores. The improvement is positive, except when α and B_{max} are both small, i.e. when the events are close to being certain and there is very little heterogeneity in the number of potential outcomes of the events.**

be the most helpful. In addition, the improvement in the correlation metric increases with the heterogeneity of the number of outcomes of the events, highlighting the benefits of our fair skill adjustment when experts forecast events with different numbers of outcomes.

In almost all cases considered in Figure 4 the fair skill adjustment improves the correlation coefficient between the experts' sample and long-term scores. Here the magnitude of ΔCorr ranges between -0.0003 and 0.03. Even larger improvements occur outside the range of parameter values considered here. For instance, if we increase B_{max} to 20, then the magnitude of ΔCorr ranges between -0.0003 and 0.05. The improvement, however, can be slightly negative, as illustrated by the black region in the left-most panels in Figure 4. Intuitively, when a large number of experts are well-informed (I_{max} is large) while the events lack inherent uncertainty (α and B_{max} are small), the "added forecast difficulty" due to an increase in the number of outcomes is almost negligible. Therefore, in this rare situation, the fair skill adjustment does not necessarily make the scores more efficient measures of the experts' forecasting skills or improve the correlation coefficient between the experts' sample and long-term scores.

In Appendix C, we repeat the analysis but under different sample sizes, namely $N = 20$ and $N = 50$. The results are qualitatively similar to those shown in Figure 4 for $N = 10$. In addition, at these sample sizes ($N \leqslant 50$), we see that the improvement in the Pearson correlation increases with the number of forecasts in the sample, from $N = 10$ to 20 and 50. However, the improvement is expected to vanish in the limit because if the number of sample forecasts is infinite, both the sample Brier score and the sample fair skill Brier score would capture experts' true forecasting skills exactly.

To check the robustness of our results against our choice of the correlation metric, Appendix D considers the Spearman correlation instead of the Pearson correlation. This metric evaluates the correlation between the ranks of the experts' sample scores and the ranks of their long-term scores. Given that the Spearman correlation is less tractable than the Pearson correlation, we use simulation to assess the improvements in the Spearman correlation coefficient. Overall, we find that these improvements align well with those discussed in this section under the Pearson correlation metric.

4. Empirical Evaluation of Fair Skill Adjustment

This section empirically explores the potential of our fair skill adjustment on data collected during two studies by the Intelligence Research Advanced Projects Activity (IARPA). The first study is the Forecasting Counterfactuals in Uncontrolled Settings (FOCUS) program[10] that was launched in 2019. The second dataset was collected by the Good Judgment Project (GJP)[11] which was one of the competing teams in IARPA's Aggregative Contingent Estimation (ACE) program[12] during 2011-2015.

4.1. Counterfactual Events

4.1.1. Data Description The FOCUS program sought to improve and evaluate answers to counterfactual questions of the form "how would history have been different if actor X had instead done action Y on date Z?". Given that we cannot re-run history and hence observe counterfactual outcomes, the program used complex simulations to generate events with different numbers of outcomes and underlying probabilities across various domains, including geopolitics, city building, and fantasy world. In particular, one simulator was based on a turn-based strategy game called Civilization V, and another was based on a variant of chess known as Reconnaissance Blind Chess. By re-running a simulation many times, we can tabulate the frequencies of different outcomes and

[10] Program website: https://www.iarpa.gov/research-programs/focus-2.

[11] The data can be downloaded at https://dataverse.harvard.edu/dataverse/gjp.

[12] Program website: https://www.iarpa.gov/research-programs/ace.

	Phase 1	Phase 2
Total # of experts	41	321
Total # of events	73	432
Average [min, max] # of forecasts per expert	35 [10, 61]	14 [10, 32]
Average [min, max] # of potential outcomes per event	5 [2, 10]	4 [2, 6]
Average [min, max] level of uncertainty of an event	0.41 [0, 0.81]	0.37 [0, 0.75]

Table 1 **Summary statistics of the events and forecasts from Phases 1 and 2 of the FOCUS program.**

construct the underlying probabilities of different outcomes (i.e., $\boldsymbol{p}_B$ in our earlier notation) of the events.

During the first 18 months of the FOCUS program (known as Phase 1), experts worked in teams to develop and refine their counterfactual forecasting methods through iterative feedback. The final forecasting methods were then handed off to a different set of expert teams that, in the next nine months of the program (known as Phase 2), made probabilistic forecasts of unprecedented counterfactual events generated by the same simulators as in Phase 1. Each team employs multiple forecasting methods to generate forecasts for the same event. In our analysis, we treat each team-method pair as a single "expert". To ensure a meaningful amount of data per expert, our analysis included only experts who made forecasts for at least ten different events. Given that the experts in the program are diverse in age, gender, race, ethnicity, income, and educational background, the experts are likely to have different levels of forecasting skills.

Table 1 provides data statistics from Phases 1 and 2. There were around 505 events in total. Our analysis treats all outcomes as categorical, regardless of whether the events were categorical or ordinal. Although Phase 1 had fewer events, these events exhibited, on average, a greater number and broader range of potential outcomes compared to Phase 2 events. The bottom row of Table 1 describes the uncertainty in the events. Here the uncertainty of an event with B potential outcomes is given by $\sum_{b=1}^{B} p_b(1-p_b)$, where p_b is the underlying probability of the bth outcome. This score ranges between 0 (lowest uncertainty) and 1 (highest uncertainty). On average, the events in neither phase are close to being certain; the events in Phase 1 are inherently slightly more uncertain than those in Phase 2. Based on these differences and our theoretical analysis in Section 3.4, we expect larger benefits from the fair skill adjustment in Phase 1 than in Phase 2.

4.1.2. Correlation Analysis Given that we only get to observe one reality, we are typically limited to comparing the experts' forecasts of real-world events against the outcomes that happen. In the FOCUS program, however, we have knowledge of the underlying probabilities of the outcomes. This puts us in a unique position where our evaluation is no longer limited by a single observed reality but can instead compare the experts' forecasts against all possible counterfactuals. To do this, we first draw an outcome for each event from the underlying probabilities, compare

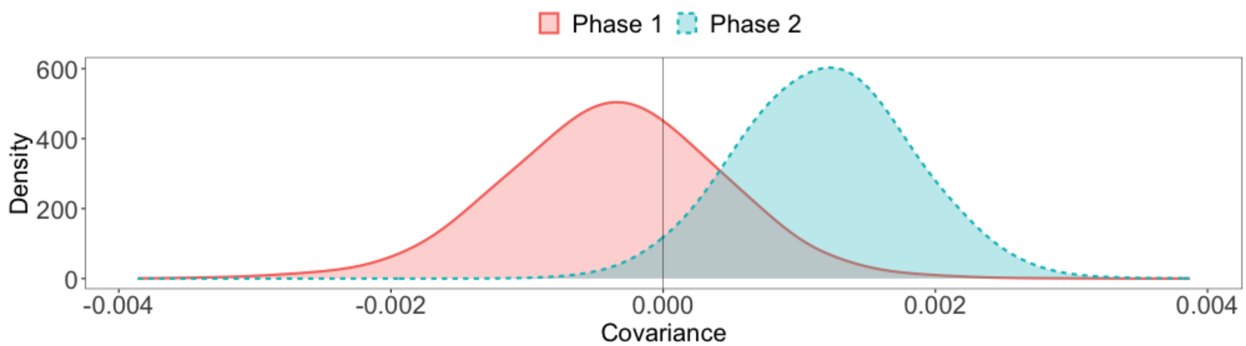

Figure 5 **The covariance between the short-term fair skill adjustment terms and the experts' long-term Brier scores.**

those outcomes against the experts' forecasts, and calculate each expert's scores. Naturally, the experts' expected (i.e., long-term) scores are not known. To approximate these expectations, we use the average of each expert's scores as their long-term score. The sample (i.e., short-term) score is calculated by randomly averaging $N = 5$ of the expert's individual scores. Here we chose to use $N = 5$ because it is the largest value that ensures that each expert's long-term score is calculated based on at least twice as many forecasts as their sample score. Experts with sample scores worse than that of the uniform forecast were excluded because such experts are not considered minimally skilled and hence are unlikely to participate in decision-making.[13] We repeat the above steps 10,000 times and arrive at 10,000 datasets of experts' sample and long-term scores.

Before looking into how our fair skill adjustment improves the correlation metric on these data, recall Section 2.2 that the fair skill adjustment can potentially improve our inference about experts' true forecasting skills if the short-term fair skill adjustment terms and the experts' long-term scores are not positively correlated. A positive covariance would indicate an unfair event assignment that favors more skilled experts. To examine whether this is the case here, Figure 5 presents the distribution of this covariance for each phase. Given that the covariance is, on average, negative in Phase 1 but positive in Phase 2, we expect a larger improvement in Phase 1.

If the covariance is non-positive, then the fair skill adjustment is expected to improve our inference about experts' true forecasting skills as long as the experts' Brier scores increase with the number of potential outcomes B at least half as fast as the scores of a uniform forecast (recall Proposition 2). Figure 6 examines this condition on the FOCUS data. It compares the distributions of the experts' sample Brier scores with the scores of the uniform forecast for varying numbers of potential outcomes B and different phases of the program. Broadly speaking, the sample Brier

[13] These experts are commonly referred to as weak learners in machine learning and do not meet the reasonable minimum requirements of an "expert".

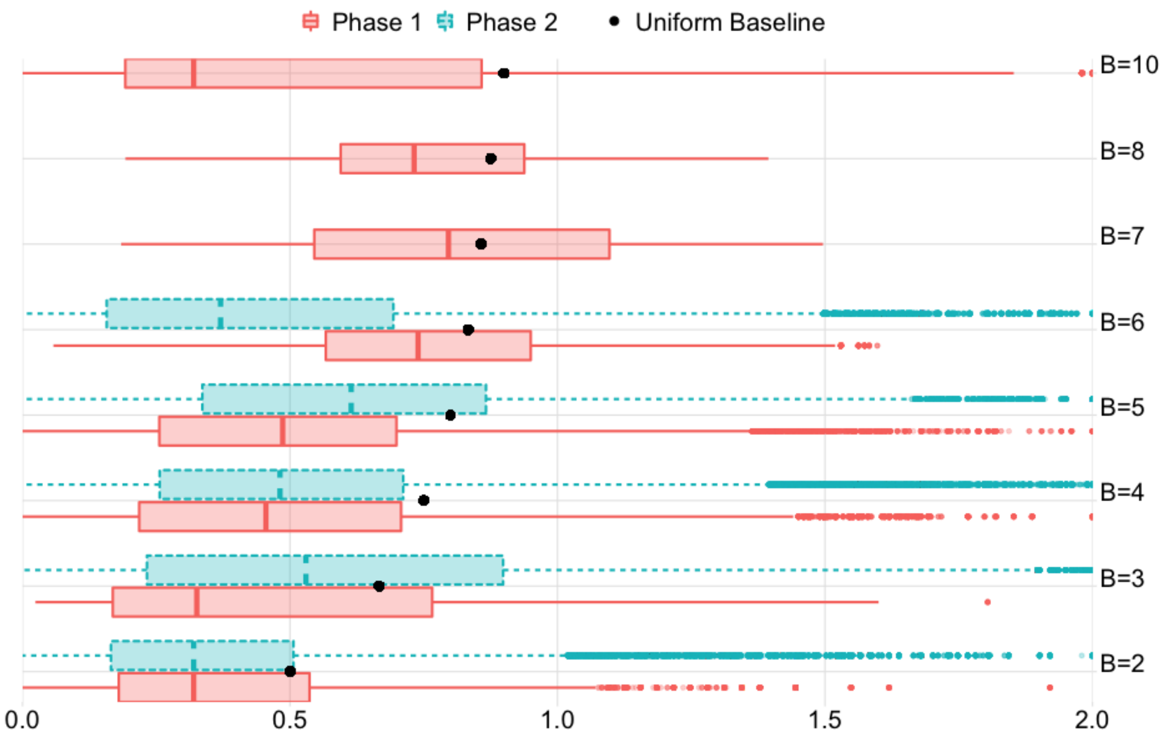

Figure 6 **The distributions of the experts' sample Brier scores conditional on the number of potential outcomes** B**.**

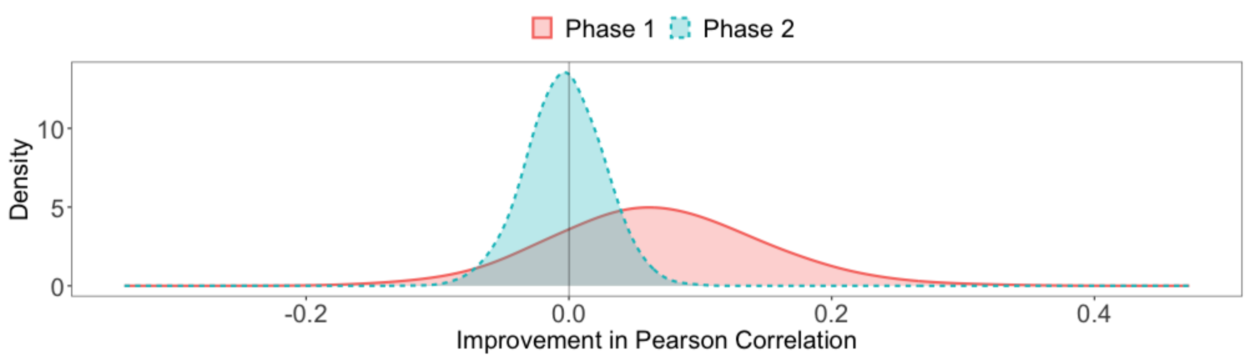

Figure 7 **The improvement in the Pearson correlation coefficient between the experts' sample and long-term scores by the fair skill adjustment.**

scores tend to increase with B faster in Phase 1 than in Phase 2. In fact, the median score in Phase 1 increases monotonically with B, except in the two most challenging cases ($B = 8$ or $B = 10$), where we also have fewer events and hence a smaller amount of data to capture typical forecasting performance. By contrast, the median score in Phase 2 zigzags from $B = 2$ to $B = 6$. Given this observation and our discussion around covariance and Figure 5, we expect to see an improvement in the correlation metric in Phase 1 but not in Phase 2.

Number of potential outcomes, B	2	3	4	5
Number of events	300	12	12	8
Average [min, max] forecast window (days)	119 [2, 549]	115 [3, 257]	86 [18, 259]	121 [6, 456]
Average [min, max] forecast horizon (days)	107 [1, 549]	112 [2, 257]	81 [9, 259]	148 [3, 456]

Table 2 Summary statistics of the events and forecast horizons in the GJP dataset.

Our speculations turn out to be correct. Figure 7 shows the distributions of improvements in our correlation metric in Phases 1 and 2. In Phase 1 the fair skill adjustment improves the correlation metric, in expectation, by around 0.07 and with probability around 0.80. The improvement observed is on par or even slightly larger than what was anticipated by our model-based analysis in Section 3. By contrast, in Phase 2 the fair skill adjustment is not expected to change the correlation metric, and we see an improvement only with a probability of around 0.45.

To check robustness, Appendix D repeats the analysis with the Spearman correlation (recall the end of Section 3). In short, the results therein resemble those presented here under the Pearson correlation: The improvement in Spearman correlation is expected to be around 0.05 and 0.01 and to occur with probabilities around 0.70 and 0.57 in Phases 1 and 2, respectively.

4.2. Future Geopolitical Events

4.2.1. Data Description The Good Judgment Project (GJP) asked thousands of experts to make probabilistic forecasts of hundreds of future geopolitical events in the ACE program. Some of these events were binary (e.g., "Would Serbia be granted European Union (EU) candidacy by December 31, 2011?" with potential outcomes "Yes" or "No"). Some were multinomial (e.g., "Who will be inaugurated as President of Russia in 2012?" with potential outcomes "Medvedev", "Putin", or "Neither"). For each event, there was a forecast window during which experts could make and update forecasts. Experts were encouraged to revise their forecasts whenever there was a change in their beliefs.[14] To keep a large number of forecast-event pairs and ensure a minimum level of uncertainty at the time of their forecast, in our analysis, we only consider each expert's most recent forecast with a forecast horizon (i.e., the number of days between the forecast date and the materialization date of the event) of at least half of the forecasting window (i.e., the number of days during which the experts can make and update their predictions) of the event. The last two rows of Table 2 describe the distributions of the forecast windows and the forecast horizons.

[14] Given that experts made forecasts at various horizons, another source of unfairness arises: Forecasting an event one day before is considerably easier than doing so hundreds of days before, introducing additional challenges in fairness evaluation. The unadjusted Brier score is affected by all these sources of heterogeneity. Our goal is to study whether and by how much the fair skill adjustment can improve our inference of experts' true forecasting skills by leveling the playing field in one particular dimension, namely the number of potential outcomes of events. Tackling other sources of heterogeneity is an important topic that is better served in a future study.

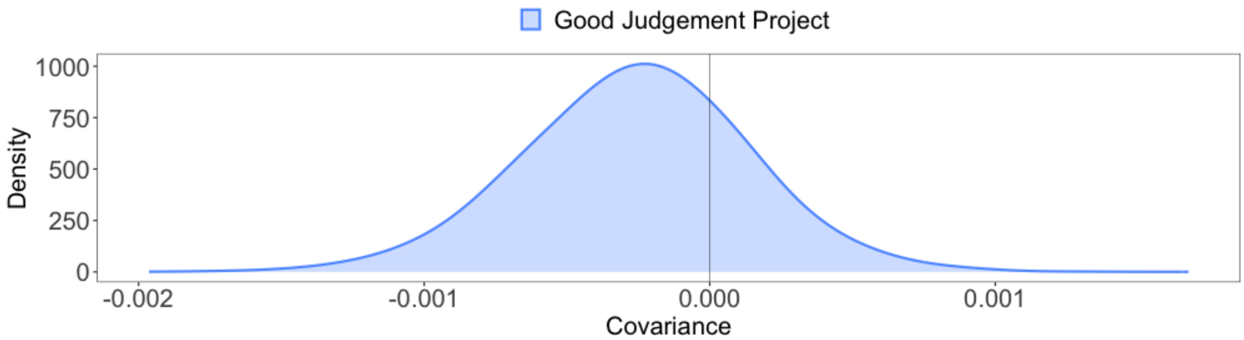

Figure 8 **The covariance between the short-term fair skill adjustment terms and the experts' long-term Brier scores.**

4.2.2. Correlation Analysis As Table 2 shows, there are 300 binary events and 32 multinomial events in our GJP dataset. To avoid highly skewed distributions of the number of potential outcomes, in each iteration of our analysis, we randomly select 11 binary events. These 11 binary events and the 32 multinomial events then form a pool of 43 categorical events. Among experts who forecast any of these events, we select those who forecast at least ten different events. As in our analysis of the FOCUS data, the average of all scores acts as a proxy for the expert's expected (long-term) score, and the average of randomly chosen $N = 5$ scores gives us the sample (short-term) score. And, as before, we exclude experts with sample scores worse than the uniform forecast because such individuals are not considered minimally skilled and are unlikely to participate in decision-making. We repeat this sampling step $10,000$ times and arrive at a total of $10,000$ datasets of experts' sample and long-term scores.

Before examining whether our fair skill adjustment improves the correlation metric on the GJP data, we form expectations based on the theory developed in Sections 2. First, Figure 8 shows the distribution of covariance between the short-term fair skill adjustment terms and the experts' long-term scores (recall condition (6) in Section 2.2). Next, Figure 9 compares the distributions of the experts' scores and the scores of the uniform forecast at different numbers of potential outcomes B (recall condition (7) in Section 2.2). Overall, the experts' scores tend to increase, albeit weakly, with the number of potential outcomes B. Given that the covariance between the experts' long-term scores and their short-term fair skill adjustment terms also tends to be negative, we expect the fair skill adjustment to improve our inference about experts' true forecasting skills. However, it is worth noting that these patterns are not as strong as those observed in Phase 1 of the FOCUS study, leading us to anticipate a positive but comparatively smaller improvement than what is illustrated in Figure 7.

Again, our theory-based speculations turn out to be correct. Figure 10 shows the distribution of improvements in our correlation metric. The fair skill adjustment improves the correlation metric

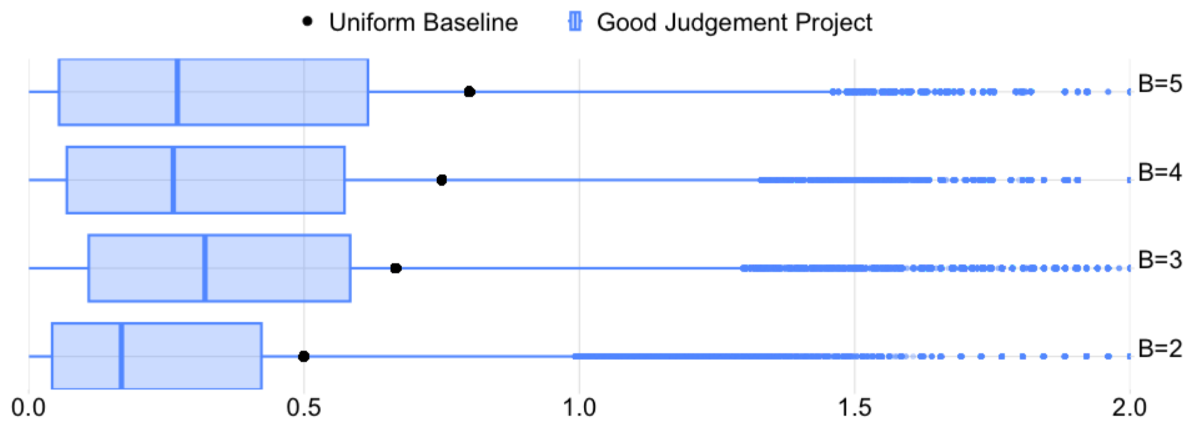

Figure 9 **The distributions of the experts' sample Brier scores conditional on the number of potential outcomes** B.

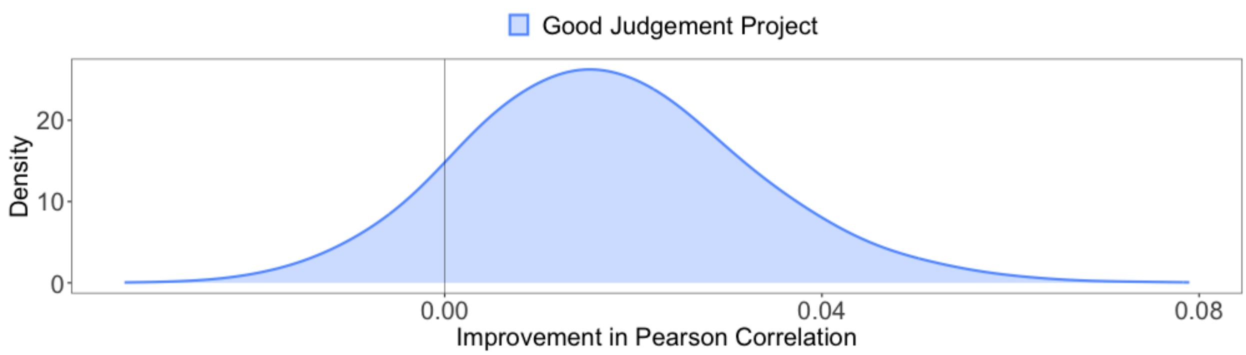

Figure 10 **The improvement in the Pearson correlation coefficient between the experts' sample and long-term scores by the fair skill adjustment.**

by approximately 0.017 on average, and with a probability of around 0.88. To check robustness, Appendix D repeats the analysis with the Spearman correlation (recall the end of Section 3) and find results that resemble those presented here under Pearson correlation: The improvement in Spearman correlation is expected to be around 0.012 and occurs with a probability of around 0.82.

4.3. Summary of Empirical Analysis

Overall, our empirical study aligns with our theoretical results. In the two datasets, the correlations between the experts' sample fair skill Brier scores and their long-term Brier scores are expected to be better or not worse than the corresponding correlations without the fair skill adjustment. Given that the adjustment is simple and can improve our inference of the experts' long-term performance, we believe it should be applied in practice whenever experts are making probabilistic forecasts of events with different numbers of outcomes.

5. Discussion

Probabilistic forecasts are frequently discussed in applications such as weather, healthcare, finance, and intelligence analysis (see O'Hagan et al. 2006, Chapter 10 for an excellent review of many published examples). The assessment of multiple experts' probabilistic forecasts should aim to eliminate chance-related discrepancies and offer dependable insights into experts' forecasting skills. This information could empower decision-makers to recognize skilled experts and leverage their forecasts to enhance decision-making in future scenarios involving unknown quantities of interest.

The Brier score is a widely employed strictly proper scoring rule to evaluate the quality of probabilistic forecasts. However, comparing experts' scores without considering the possibility that they may have forecasted events with different levels of forecastability can lead to unfair comparisons. While heterogeneity in difficulty can stem from various sources, past literature has primarily focused on the base rates of the events. Unfortunately, base rates of one-off events are rarely available, which makes it challenging to control for their effects in performance comparisons.

In this paper, we turned our attention to a previously unexplored source, namely the heterogeneity of the number of outcomes of the events. Given that the number of outcomes is known, adjusting for its effects on the inherent uncertainty is possible even in tasks involving one-off events. To this end, we proposed the fair skill adjustment. Unlike past work that has primarily used axiomatic analysis to compare scoring rules, we proposed and used the correlation coefficient between the experts' sample and long-term scores as a metric for comparing different scoring rules. Based on both theoretical (Sections 2 and 3) and empirical analyses (Section 4), we showed that the fair skill adjustment improves this correlation metric over a wide range of plausible forecasting environments and hence is better at identifying skilled experts when the experts make predictions of events with different numbers of outcomes. In addition, the fair skill adjustment is intuitive, straightforward, and applicable without requiring additional data; that is, it is accessible to practitioners whenever the Brier score is accessible. Finally, the adjustment is objective, does not rely on experts' forecasts, and does not alter experts' truth-telling behavior, as the fair skill adjusted score remains strictly proper.

Our research suggests several potential directions for future research. One direction is to explore the adjustment of the scoring rule for probabilistic forecasts of events with varying numbers of ordinal outcomes. Given that the order of outcomes is important in this context, our argument for a symmetric parametric model (see end of Section 3.2) would no longer hold directly. Either a different argument is needed or we must develop a new model that can parsimoniously capture forecasts of events with ordinal outcomes.

Another promising avenue for future research involves addressing other sources of inherent uncertainty beyond the number of potential outcomes. For instance, events occurring further into the future are typically associated with higher uncertainty levels than those happening in the near future. Considering the importance of both accuracy and timeliness in forecasting, it would be advantageous to provide extra credit to experts who can make accurate forecasts earlier. However, establishing a natural relationship between the amount of compensation and the forecast horizon poses a challenge. Therefore, it is not clear how to construct a suitable baseline for adjustments akin to the uniform baseline used in the fair skill Brier score. Other observable sources of difficulty include the specific topics or nature of the forecasted events. For instance, variables related to large institutions that change slowly may be easier to forecast than variables associated with individual human behavior, which can be more unpredictable.

While compensating for all sources of difficulty may be challenging, this paper demonstrates that even partial compensation can enhance our inference. Through this analysis, our objective is to shift the focus away from base rates, raise awareness, and encourage methodological advancements in accounting for the many other sources of heterogeneity that can hinder our ability to identify highly skilled expert forecasters.

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Appendix A: An Example of A Fairness Concern.

The following numerical example illustrates how a fairness concern could arise when the average score is used as a baseline in calculating the "relative" skill score.

EXAMPLE 1. Consider three questions and three experts.

Suppose Question 1 has a binary format, and the realized outcome is $(1,0)$. Question 2 has three potential outcomes, and the realized outcome is $(0,1,0)$. Finally, Question 3 has four potential outcomes, and the realized outcome is $(0,0,0,1)$. Expert A answers Questions 1 and 2, Expert B answers Questions 2 and 3, and Expert C answers Questions 1 and 3, their forecasts and achieved Brier scores are as follows:

- Expert A's forecast on Question 1 is $(0.6, 0.4)$, and achieves 0.32;
- Expert A's forecast on Question 2 is $(0.3, 0.4, 0.3)$, and achieves 0.54;
- Expert B's forecast on Question 2 is $(0.1, 0.4, 0.5)$, and achieves 0.62;
- Expert B's forecast on Question 3 is $(0, 0, 0.4, 0.6)$, and achieves 0.32;
- Expert C's forecast on Question 1 is $(0.9, 0.1)$, and achieves 0.02;
- Expert C's forecast on Question 3 is $(0.2, 0.3, 0.2, 0.3)$, and achieves 0.66.

Using the "relative" skill score, Expert A was deemed more skilled than Expert B $(0.54-(0.54+0.62)/2 < 0.62-(0.54+0.62)/2)$. However, after introducing Expert C to the group, their "relative" skill scores changed as follows:

- Expert A's "relative" skill score becomes $((0.54-(0.54+0.62)/2)+(0.32-(0.32+0.02)/2))/2 = (-0.04+0.15)/2 = 0.055$;
- Expert B's "relative" skill score becomes $((0.62-(0.54+0.62)/2)+(0.32-(0.32+0.66)/2))/2 = (0.04-0.17)/2 = -0.065$.

Now, Expert B is deemed more skilled than Expert A based on their "relative" skill scores $(-0.065 < 0.055)$, which is the opposite of the previous conclusion. The fair skill Brier score, on the other hand, is unaffected by Expert C's inclusion and remains consistent:

- Expert A's fair skill Brier score is $(0.32-1/2+0.54-2/3)/2 = -0.153$;
- Expert B's fair skill Brier score is $(0.62-2/3+0.32-3/4)/2 = -0.238$.

Expert B remains more skilled than Expert A based on their fair skill Brier scores $(-0.238 < -0.153)$, regardless of Expert C's participation.

Appendix B: Proofs of Proposition and Theorems

B.1. Proof of Proposition 1.

Given that the fair skill adjustment term does not depend on the expert's forecast, the expected fair skill Brier score and Brier score are minimized by the same forecast. This forecast is the expert's actual belief because the Brier score is strictly proper (e.g., Gneiting and Raftery 2007). Therefore, the fair skill Brier score is also strictly proper. □

B.2. Proof of Proposition 2.

Suppose the number of outcomes B follows some discrete distribution π_B. Then,

$$\begin{aligned}
&\mathrm{Var}\left(\mathbb{E}\left[\mathrm{fsBrS}_k|B\right]\right) < \mathrm{Var}\left(\mathbb{E}\left[\mathrm{BrS}_k|B\right]\right)\\
\Leftrightarrow\ &\mathrm{Var}\left(\mathbb{E}\left[\mathrm{BrS}_k|B\right]\right) - 2\mathrm{Cov}\left(\mathbb{E}\left[\mathrm{BrS}_k|B\right], \mathbb{E}\left[\mathrm{U}_k|B\right]\right) + \mathrm{Var}\left(\mathbb{E}\left[\mathrm{U}_k|B\right]\right) < \mathrm{Var}\left(\mathbb{E}\left[\mathrm{BrS}_k|B\right]\right)\\
\Leftrightarrow\ &\mathrm{Cov}\left(\mathbb{E}\left[\mathrm{BrS}_k|B\right], \mathbb{E}\left[\mathrm{U}_k|B\right]\right) > \mathrm{Var}\left(\mathbb{E}\left[\mathrm{U}_k|B\right]\right)/2\\
\Leftrightarrow\ &\mathrm{Cov}\left(\mathbb{E}\left[\mathrm{BrS}_k|B\right] - \mathbb{E}\left[\mathrm{U}_k|B\right]/2, \mathbb{E}\left[\mathrm{U}_k|B\right]\right) > 0.
\end{aligned}$$

Given that $\mathbb{E}\left[\mathrm{U}_k|B\right] = (B-1)/B$ increases with B monotonically, $\mathrm{Cov}\left(\mathbb{E}\left[\mathrm{BrS}_k|B\right] - \mathbb{E}\left[\mathrm{U}_k|B\right]/2, \mathbb{E}\left[\mathrm{U}_k|B\right]\right)$ is positive for all π_B if and only if $\mathbb{E}\left[\mathrm{BrS}_k|B\right] - \mathbb{E}\left[\mathrm{U}_k|B\right]/2$ increases with B monotonically (Thorisson 1995), that is, if for all B we have that $\mathbb{E}\left[\mathrm{BrS}_k|(B+1)\right] - \mathbb{E}\left[\mathrm{BrS}_k|B\right] > 0.5\left[B/(B+1) - (B-1)/B\right]$. $\square$

B.3. Proof of Proposition 3.

Given that $\mathbb{E}[p_b] = \alpha/(B\alpha) = 1/B$ and $\mathrm{Var}(p_b) = \alpha(B\alpha-\alpha)/(B^2\alpha^2(B\alpha+1)) = (B-1)/(B^2(B\alpha+1))$, we have that $\mathbb{E}[p_b^2] = \mathrm{Var}(p_b) + \mathbb{E}[p_b]^2 = (1+\alpha)/(B(B\alpha+1))$. Plugging this into (2) gives us $1 - \sum_{b=1}^{B}\mathbb{E}[p_b^2] = 1-(1+\alpha)/(1+B\alpha)$. $\square$

B.4. Proof of Proposition 4.

By Proposition 2, we have that $\mathrm{Var}\left(\mathbb{E}\left[\mathrm{fsBrS}_{\text{perfect-skill}}|B\right]\right) < \mathrm{Var}\left(\mathbb{E}\left[\mathrm{BrS}_{\text{perfect-skill}}|B\right]\right)$ if

$$\begin{aligned}
&\mathbb{E}\left[\mathrm{BrS}_{\text{perfect-skill}}|(B+1)\right] - \mathbb{E}\left[\mathrm{BrS}_{\text{perfect-skill}}|B\right] > 0.5\left[\frac{B}{B+1} - \frac{B-1}{B}\right]\\
\Leftrightarrow &\left[1 - \frac{1+\alpha}{1+(B+1)\alpha}\right] - \left[1 - \frac{1+\alpha}{1+B\alpha}\right] > \frac{1}{2B(B+1)}\\
\Leftrightarrow\ &2B(B+1)(1+\alpha)\left[\frac{1}{1+B\alpha} - \frac{1}{1+(B+1)\alpha}\right] > 1.
\end{aligned} \tag{16}$$

The derivative of the left-hand side of (16) with respect to B is

$$\frac{2\alpha(1+\alpha)(1+\alpha+2B+2\alpha B+2\alpha B^2)}{(1+\alpha B)^2(1+\alpha+\alpha B)^2} > 0.$$

Therefore, the left-hand side of (16) increases with B monotonically. Substituting the smallest value $B=2$ into (16) then gives

$$\begin{aligned}
&12(1+\alpha)\left[\frac{1}{1+2\alpha} - \frac{1}{1+3\alpha}\right] > 1\\
&\Leftrightarrow \alpha > \left(\sqrt{73}-7\right)/12 \approx 0.13.
\end{aligned}$$

$\square$

B.5. Proof of Proposition 5.

Conditional on $\boldsymbol{p}_B$, the expected Brier score of $\boldsymbol{f}_B$ is

$$\begin{aligned}
\mathbb{E}\left[\mathrm{BrS}_k|\boldsymbol{p}_B\right] &= \mathbb{E}\left[\boldsymbol{f}_B^2\right] - 2\mathbb{E}\left[\boldsymbol{f}_B\right]\boldsymbol{p}_B + 1\\
&= \left(\mathbb{E}\left[\boldsymbol{f}_B\right] - \boldsymbol{p}_B\right)^2 + \mathbb{E}\left[\boldsymbol{f}_B^2\right] - \mathbb{E}\left[\boldsymbol{f}_B\right]^2 + \mathbb{E}\left[\boldsymbol{y}_B^2\right] - \mathbb{E}\left[\boldsymbol{y}_B\right]^2.
\end{aligned} \tag{17}$$

Then, by linearity and Ouimet (2021),

$$\begin{aligned}
\left(\mathbb{E}\left[\boldsymbol{f}_B\right]-\boldsymbol{p}_B\right)^2 &= \sum_{b=1}^{B}\left(\mathbb{E}\left[\frac{\beta+x_b}{B\beta+I}\right]-p_b\right)^2 \\
&= \sum_{b=1}^{B}\left(\frac{\beta+\mathbb{E}\left[x_b\right]}{B\beta+I}-p_b\right)^2 \\
&= \sum_{b=1}^{B}\left(\frac{\beta+Ip_b}{B\beta+I}-p_b\right)^2 \\
&= \left(\frac{B\beta}{B\beta+I}\right)^2\left(\sum_{b=1}^{B}p_b^2-\frac{1}{B}\right), \\
\mathbb{E}\left[\boldsymbol{f}_B^2\right]-\mathbb{E}\left[\boldsymbol{f}_B\right]^2 &= \mathbb{E}\left[\sum_{b=1}^{B}\left(\frac{\beta+x_b}{B\beta+I}\right)^2\right]-\sum_{b=1}^{B}\left(\mathbb{E}\left[\frac{\beta+x_b}{B\beta+I}\right]\right)^2 \\
&= \sum_{b=1}^{B}\frac{\beta^2+2\beta\mathbb{E}\left[x_b\right]+\mathbb{E}\left[x_b^2\right]}{\left(B\beta+I\right)^2}-\sum_{b=1}^{B}\left(\frac{\beta+\mathbb{E}\left[x_b\right]}{B\beta+I}\right)^2 \\
&= \sum_{b=1}^{B}\frac{\beta^2+2\beta Ip_b+\left(I+I\left(I-1\right)p_b\right)p_b}{\left(B\beta+I\right)^2}-\sum_{b=1}^{B}\left(\frac{\beta+Ip_b}{B\beta+I}\right)^2 \\
&= \frac{I}{\left(B\beta+I\right)^2}\left(1-\sum_{b=1}^{B}p_b^2\right), \text{ and} \\
\mathbb{E}\left[\boldsymbol{y}_B^2\right]-\mathbb{E}\left[\boldsymbol{y}_B\right]^2 &= 1-\sum_{b=1}^{B}p_b^2.
\end{aligned}$$

Plugging these into (17) gives

$$\mathbb{E}\left[\mathrm{BrS}_k|\boldsymbol{p}_B\right]=\left(\frac{B\beta}{B\beta+I}\right)^2\left(\sum_{b=1}^{B}p_b^2-\frac{1}{B}\right)+\frac{I}{\left(B\beta+I\right)^2}\left(1-\sum_{b=1}^{B}p_b^2\right)+\left(1-\sum_{b=1}^{B}p_b^2\right).$$

Taking expectations with respect to $\boldsymbol{p}_B$ gives the desired form of the expected Brier score:

$$\mathbb{E}\left[\mathrm{BrS}_k\right]=\left(\frac{B\beta}{B\beta+I}\right)^2\left(\mathbb{E}\left[\sum_{b=1}^{B}p_b^2\right]-\frac{1}{B}\right)+\frac{I}{\left(B\beta+I\right)^2}\left(1-\mathbb{E}\left[\sum_{b=1}^{B}p_b^2\right]\right)+\left(1-\mathbb{E}\left[\sum_{b=1}^{B}p_b^2\right]\right),$$

where

$$\mathbb{E}\left[\sum_{b=1}^{B}p_b^2\right]=\sum_{b=1}^{B}\mathbb{E}\left[p_b^2\right]=\sum_{b=1}^{B}\frac{\alpha\left(\alpha+1\right)}{B\alpha\left(B\alpha+1\right)}=\frac{B\alpha\left(\alpha+1\right)}{B\alpha\left(B\alpha+1\right)}=\frac{\alpha+1}{B\alpha+1}.$$

It follows that

$$\mathbb{E}\left[\mathrm{BrS}_k|\alpha,\beta,B,I\right]=\frac{B-1}{B\alpha+1}\left(\frac{B\beta^2+I\alpha}{\left(B\beta+I\right)^2}+\alpha\right).$$

Finally, observe that

$$\begin{aligned}
&\frac{B-1}{B\alpha+1}\left(\frac{B\beta^2+I\alpha}{\left(B\beta+I\right)^2}+\alpha\right)<\frac{B-1}{B} \\
&\qquad\Leftrightarrow \frac{B\beta^2+I\alpha}{\left(B\beta+I\right)^2}+\alpha<\frac{B\alpha+1}{B}=\alpha+\frac{1}{B} \\
&\qquad\Leftrightarrow B\beta^2+I\alpha<\frac{B^2\beta^2+2B\beta I+I^2}{B} \\
&\qquad\Leftrightarrow B\beta^2+I\alpha<B\beta^2+2\beta I+I^2/B \\
&\qquad\Leftrightarrow \alpha<2\beta+I/B \text{ when } I>0; \\
&\frac{B-1}{B\alpha+1}\left(\frac{B\beta^2+I\alpha}{\left(B\beta+I\right)^2}+\alpha\right)=\frac{B-1}{B} \text{ when } I=0.
\end{aligned}$$

□

B.6. Derivation of ΔCorr.

Recall (4) that the correlation between the experts' short-term and long-term scores is

$$\mathrm{Corr}\left(\overline{S}, \mathbb{E}[S]\right) = \frac{\mathrm{Cov}\left(\overline{S}, \mathbb{E}[S]\right)}{\sqrt{\mathrm{Var}\left(\overline{S}\right)\mathrm{Var}\left(\mathbb{E}[S]\right)}}.$$

It follows that

$$\mathrm{Corr}\left(\overline{\mathrm{BrS}}, \mathbb{E}[\mathrm{BrS}]\right) = \frac{\mathrm{Cov}\left(\overline{\mathrm{BrS}}, \mathbb{E}[\mathrm{BrS}]\right)}{\sqrt{\mathrm{Var}\left(\overline{\mathrm{BrS}}\right)\mathrm{Var}\left(\mathbb{E}[\mathrm{BrS}]\right)}}. \tag{18}$$

Suppose β is the same for all experts and α for all events. Let I be randomly drawn from a discrete distribution π_I and B be randomly drawn from a discrete distribution π_B. By linearity, $\mathrm{Cov}\left(\overline{\mathrm{BrS}}, \mathbb{E}[\mathrm{BrS}]\right) = \mathrm{Cov}\left(\mathrm{BrS}, \mathbb{E}[\mathrm{BrS}]\right)$. By the law of large numbers, $\mathrm{Cov}\left(\mathrm{BrS}, \mathbb{E}[\mathrm{BrS}]\right) = \mathrm{Var}\left(\mathbb{E}[\mathrm{BrS}]\right)$ as the number of experts approaches infinity.

$$\begin{aligned}
\mathrm{Var}\left(\mathbb{E}[\mathrm{BrS}]\right) &= \int \mathbb{E}[\mathrm{BrS}|I]^2\, \pi_I \mathrm{d}I - \left(\int \mathbb{E}[\mathrm{BrS}|\, I]\pi_I \mathrm{d}I\right)^2, \text{ where} \\
\mathbb{E}[\mathrm{BrS}|I] &= \int \frac{B-1}{B\alpha+1}\left(\frac{B\beta^2+I\alpha}{(B\beta+I)^2}+\alpha\right)\pi_B \mathrm{d}B.
\end{aligned}$$

By the law of total variance,

$$\begin{aligned}
\mathrm{Var}\left(\overline{\mathrm{BrS}}\right) &= \underbrace{\mathrm{Var}\left(\mathbb{E}\left[\overline{\mathrm{BrS}}\right]\right)}_{\text{Cross-Component Variation}} + \underbrace{\mathbb{E}\left[\mathrm{Var}\left(\overline{\mathrm{BrS}}\right)\right]}_{\text{Within-Component Variation}} \\
&= \mathrm{Var}\left(\mathbb{E}[\mathrm{BrS}]\right) + \int \frac{1}{N}\mathrm{Var}\left(\mathrm{BrS}|I\right)\pi_I \mathrm{d}I, \text{ where} \\
\mathrm{Var}\left(\mathrm{BrS}|I\right) &= \mathbb{E}\left[\mathrm{BrS}^2|I\right] - \mathbb{E}[\mathrm{BrS}|I]^2 = \int \mathbb{E}\left[\mathrm{BrS}^2|B,I\right]\pi_B \mathrm{d}B - \mathbb{E}[\mathrm{BrS}|I]^2.
\end{aligned}$$

Accordingly,

$$\mathrm{Corr}\left(\overline{\mathrm{BrS}}, \mathbb{E}[\mathrm{BrS}]\right) = \frac{\mathrm{Cov}\left(\overline{\mathrm{BrS}}, \mathbb{E}[\mathrm{BrS}]\right)}{\sqrt{\mathrm{Var}\left(\overline{\mathrm{BrS}}\right)\mathrm{Var}\left(\mathbb{E}[\mathrm{BrS}]\right)}} = \sqrt{\frac{\mathrm{Var}\left(\mathbb{E}[\mathrm{BrS}]\right)}{\mathrm{Var}\left(\mathbb{E}[\mathrm{BrS}]\right) + \mathbb{E}\left[\mathrm{Var}\left(\overline{\mathrm{BrS}}\right)\right]}}.$$

The expert's forecasts about the outcomes are

$$\boldsymbol{f}_B = \frac{\boldsymbol{\beta}_B + \boldsymbol{x}_B}{B\beta+I},$$

where $\boldsymbol{\beta}_B = (\beta, \ldots, \beta)$ and $\boldsymbol{x}_B \sim \mathrm{Multinomial}(\boldsymbol{p}_B, I)$. Conditional on $\boldsymbol{p}_B$, the second moment of the expert's Brier score is

$$\begin{aligned}
\mathbb{E}\left[\mathrm{BrS}^2|\boldsymbol{p}_B\right] &= \sum_{b^*=1}^{B} p_{b^*}\left(1 - 2f_{b^*} + \sum_{b=1}^{B} f_b^2\right)^2 \\
&= \left(1 + \sum_{b=1}^{B} f_b^2\right)^2 + 4\sum_{b=1}^{B} p_b f_b^2 - 4\left(1 + \sum_{b=1}^{B} f_b^2\right)\sum_{b=1}^{B} p_b f_b,
\end{aligned}$$

where $\boldsymbol{p}_B \sim \text{Dirichlet}(\boldsymbol{\alpha}_B)$ and $\boldsymbol{\alpha}_B = (\alpha, \dots, \alpha)$. By Ouimet (2021) and linearity,

$$\begin{aligned}
&\mathbb{E}\left[\mathrm{BrS}^2|\alpha,\beta,B,I\right] \\
&= -1 + 2\left(\frac{B-1}{B\alpha+1}\left(\alpha + \frac{I\alpha+B\beta}{(I+B\beta)^2}\right)\right) \\
&\quad + \frac{4}{(I+B\beta)^2}\left(\beta^2 + \frac{I}{B\alpha+1}(\alpha+1)(2\beta+1) + \frac{I(I-1)}{(B\alpha+1)(B\alpha+2)}(\alpha+1)(\alpha+2)\right) \\
&\quad - \frac{4}{(I+B\beta)^3}\Bigg(B\beta^3 + \frac{I}{B\alpha+1}\left((2\beta+1)(\alpha+\beta+1) + B\beta(\alpha+\beta+3\alpha\beta)\right) + \\
&\qquad\qquad \frac{I(I-1)}{(B\alpha+1)(B\alpha+2)}\left(2(\alpha+1)(\alpha+3\beta+3) + B\alpha(\alpha+1)(3\beta+1)\right) + \\
&\qquad\qquad \frac{I(I-1)(I-2)}{(B\alpha+1)(B\alpha+2)(B\alpha+3)}\left(2(\alpha+1)(2\alpha+3) + B\alpha(\alpha+1)^2\right)\Bigg) \\
&\quad + \frac{1}{(I+B\beta)^4}\Bigg(B^2\beta^4 + I\left(2B\beta^2(2\beta+1) + (2\beta+1)^2\right) + \\
&\qquad\qquad \frac{I(I-1)}{B\alpha+1}\left((B\alpha+1)(2\beta+1)^2 + 2B\beta^2(\alpha+1) + (8\beta+6)(\alpha+1)\right) + \\
&\qquad\qquad \frac{I(I-1)(I-2)}{(B\alpha+1)(B\alpha+2)}\left(2B\alpha(\alpha+1)(2\beta+1) + 4(\alpha+1)(\alpha+2\beta+3)\right) + \\
&\qquad\qquad \frac{I(I-1)(I-2)(I-3)}{(B\alpha+1)(B\alpha+2)(B\alpha+3)}\left(B\alpha(\alpha+1)^2 + 2(\alpha+1)(2\alpha+3)\right)\Bigg).
\end{aligned}$$

Then, we can write $\mathrm{Corr}\left(\overline{\mathrm{BrS}}, \mathbb{E}[\mathrm{BrS}]\right)$ in terms of α, β, B, I, and N accordingly.

Akin to (18),

$$\mathrm{Corr}\left(\overline{\mathrm{fsBrS}}, \mathbb{E}[\mathrm{fsBrS}]\right) = \frac{\mathrm{Cov}\left(\overline{\mathrm{fsBrS}}, \mathbb{E}[\mathrm{fsBrS}]\right)}{\sqrt{\mathrm{Var}\left(\overline{\mathrm{fsBrS}}\right)\mathrm{Var}(\mathbb{E}[\mathrm{fsBrS}])}} = \sqrt{\frac{\mathrm{Var}(\mathbb{E}[\mathrm{fsBrS}])}{\mathrm{Var}(\mathbb{E}[\mathrm{fsBrS}]) + \mathbb{E}\left[\mathrm{Var}\left(\overline{\mathrm{fsBrS}}\right)\right]}}.$$

$$\begin{aligned}
\mathrm{Var}(\mathbb{E}[\mathrm{fsBrS}]) &= \int \mathbb{E}[\mathrm{fsBrS}|I]^2\,\pi_I \mathrm{d}I - \left(\int \mathbb{E}[\mathrm{fsBrS}|\,I]\pi_I \mathrm{d}I\right)^2, \text{ where} \\
\mathbb{E}[\mathrm{fsBrS}|I] &= \int \frac{B-1}{B\alpha+1}\left(\frac{B\beta^2+I\alpha}{(B\beta+I)^2} + \alpha\right) - \frac{B-1}{B}\pi_B \mathrm{d}B.
\end{aligned}$$

$$\begin{aligned}
&\mathbb{E}\left[\mathrm{Var}\left(\overline{\mathrm{fsBrS}}\right)\right] = \int \frac{1}{N}\mathrm{Var}(\mathrm{fsBrS}|I)\,\pi_I \mathrm{d}I, \text{ where} \\
&\quad \mathrm{Var}(\mathrm{fsBrS}|I) = \mathbb{E}\left[\mathrm{fsBrS}^2|I\right] - \mathbb{E}[\mathrm{fsBrS}|I]^2 = \int \mathbb{E}\left[\mathrm{fsBrS}^2|B,I\right]\pi_B \mathrm{d}B - \mathbb{E}[\mathrm{fsBrS}|I]^2.
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}\left[\mathrm{fsBrS}^2|\boldsymbol{p}_B\right] &= \sum_{b^*=1}^{B} p_{b^*}\left(1 - 2f_{b^*} + \sum_{b=1}^{B} f_b^2 - \frac{B-1}{B}\right)^2 \\
&= \mathbb{E}\left[\mathrm{BrS}^2|\boldsymbol{p}_B\right] - 2\left(\frac{B-1}{B}\right)\mathbb{E}[\mathrm{BrS}|\boldsymbol{p}_B] + \left(\frac{B-1}{B}\right)^2.
\end{aligned}$$

It follows that

$$\mathbb{E}\left[\mathrm{fsBrS}^2|\alpha,\beta,B,I\right]$$

$$
\begin{aligned}
=&\mathbb{E}\left[\mathrm{BrS}^2|\alpha,\beta,B,I\right]-2\left(\frac{B-1}{B}\right)\mathbb{E}\left[\mathrm{BrS}|\alpha,\beta,B,I\right]+\left(\frac{B-1}{B}\right)^2\\
=&-1+2\left(\frac{B-1}{B\alpha+1}\left(\alpha+\frac{I\alpha+B\beta}{(I+B\beta)^2}\right)\right)\\
&+\frac{4}{(I+B\beta)^2}\left(\beta^2+\frac{I}{B\alpha+1}(\alpha+1)(2\beta+1)+\frac{I(I-1)}{(B\alpha+1)(B\alpha+2)}(\alpha+1)(\alpha+2)\right)\\
&-\frac{4}{(I+B\beta)^3}\Bigg(B\beta^3+\frac{I}{B\alpha+1}\left((2\beta+1)(\alpha+\beta+1)+B\beta(\alpha+\beta+3\alpha\beta)\right)+\\
&\qquad\frac{I(I-1)}{(B\alpha+1)(B\alpha+2)}\left(2(\alpha+1)(\alpha+3\beta+3)+B\alpha(\alpha+1)(3\beta+1)\right)+\\
&\qquad\frac{I(I-1)(I-2)}{(B\alpha+1)(B\alpha+2)(B\alpha+3)}\left(2(\alpha+1)(2\alpha+3)+B\alpha(\alpha+1)^2\right)\Bigg)\\
&+\frac{1}{(I+B\beta)^4}\Bigg(B^2\beta^4+I\left(2B\beta^2(2\beta+1)+(2\beta+1)^2\right)+\\
&\qquad\frac{I(I-1)}{B\alpha+1}\left((B\alpha+1)(2\beta+1)^2+2B\beta^2(\alpha+1)+(8\beta+6)(\alpha+1)\right)+\\
&\qquad\frac{I(I-1)(I-2)}{(B\alpha+1)(B\alpha+2)}\left(2B\alpha(\alpha+1)(2\beta+1)+4(\alpha+1)(\alpha+2\beta+3)\right)+\\
&\qquad\frac{I(I-1)(I-2)(I-3)}{(B\alpha+1)(B\alpha+2)(B\alpha+3)}\left(B\alpha(\alpha+1)^2+2(\alpha+1)(2\alpha+3)\right)\Bigg)\\
&-2\left(\frac{B-1}{B}\right)\frac{B-1}{B\alpha+1}\left(\frac{B\beta^2+I\alpha}{(B\beta+I)^2}+\alpha\right)+\left(\frac{B-1}{B}\right)^2.
\end{aligned}
$$

Then, we can write $\mathrm{Corr}\left(\overline{\mathrm{fsBrS}},\mathbb{E}\left[\mathrm{fsBrS}\right]\right)$ in terms of α, β, B, I, and N accordingly. Recall (5) that $\mathrm{Corr}\left(\overline{\mathrm{fsBrS}},\mathbb{E}\left[\mathrm{fsBrS}\right]\right)=\mathrm{Corr}\left(\overline{\mathrm{fsBrS}},\mathbb{E}\left[\mathrm{BrS}\right]\right)$. Then, we can write $\Delta\mathrm{Corr}=\mathrm{Corr}\left(\overline{\mathrm{fsBrS}},\mathbb{E}\left[\mathrm{BrS}\right]\right)-\mathrm{Corr}\left(\overline{\mathrm{BrS}},\mathbb{E}\left[\mathrm{BrS}\right]\right)$ in terms of α, β, B, I, and N.

Appendix C: Additional Illustrations

To provide additional illustrations to Figure 4, in which the number of sample forecasts is fixed to be $N=10$, we include two more cases here ($N=20$ and $N=50$).

Figure 11 fixes the number of sample forecasts to $N=20$, and Figure 12 fixes the number of sample forecasts to $N=50$. The rest of the parameters are the same as those in Figure 4. Both Figure 11 and Figure 12 show similar patterns as those in Figure 4. In addition, at these low sample sizes ($N\leqslant 50$), the improvement in the Pearson correlation increases with the number of forecasts in the sample, from $N=10$ to 20 and 50. However, the improvement is expected to vanish at the limit where the number of sample forecasts is infinite because then the sample Brier score and the sample fair skill Brier score would both capture experts' true forecasting skills.

Appendix D: Robustness Checks

As a robustness check, we plot the Spearman versions of Figure 4, Figure 11, and Figure 12. Unlike the Pearson correlation, it is impossible to derive a closed-form expression for $\Delta\mathrm{Corr}$ of the Spearman version. Therefore, each pixel of Figure 13, Figure 14, and Figure 15 is based on simulated results. In our simulations, each forecast is compared against a materialized outcome, and a score is then generated. The behavioral

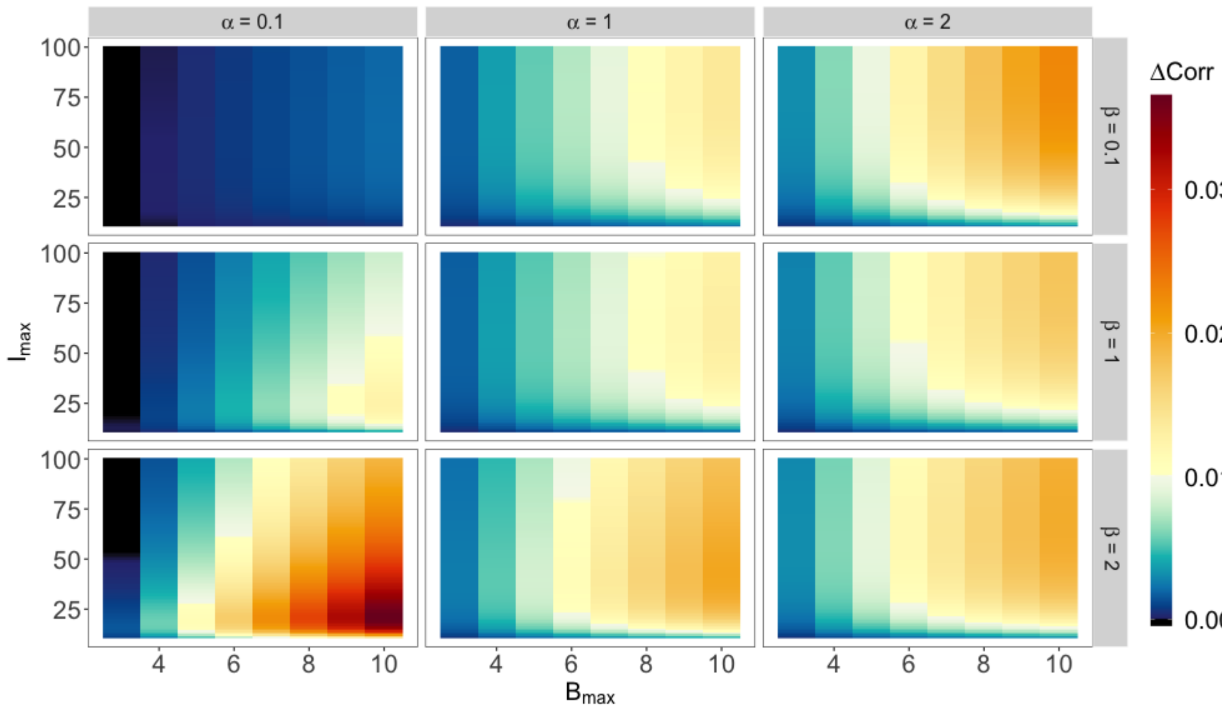

Figure 11 **The amounts by which fair skill adjustment improves the Pearson correlation between the experts' sample ($N = 20$) and long-term scores. The improvement is positive, except when α and $B_{\max}$ are both small, i.e., when the events are close to being certain, and there is very little heterogeneity in the number of potential outcomes of the events.**

model of experts' forecasts behind the simulations is consistent with our construction in Section 3. In each pixel, we fix the number of long-term forecasts to 10,000, the minimum level of information to $I_{\min} = 10$, and the minimum number of outcomes to $B_{\min} = 2$. Conditional on α, β, $I_{\max}$, and $B_{\max}$, we first compute 10,000 experts' long-term scores, assuming their level of information being uniformly distributed between $I_{\min}$ and $I_{\max}$ and the number of outcomes of the events being uniformly distributed between $B_{\min}$ and $B_{\max}$. Then, we fix the number of sample forecasts to $N = 10$ (Figure 13), $N = 20$ (Figure 14), and $N = 50$ (Figure 15), and compute these 10,000 experts' sample scores based on each expert's sample forecasts. At last, we compute the improvement in the Spearman correlation by the fair skill adjustment and complete the calculation of one pixel.

Overall, we find that Figure 13, Figure 14, and Figure 15 are consistent with their corresponding Pearson versions (Figure 4, Figure 11, and Figure 12, respectively). The patterns in Figure 13, Figure 14, and Figure 15 hence exhibit the robustness of the improvement in correlations by the fair skill adjustment. In addition, since the Spearman correlation measures the correlation between the ranks of the sample scores and the ranks of the long-term scores, these patterns indicate that the fair skill adjustment improves the decision-maker's ability to identify skilled experts, even if they are only interested in the ranking of experts.

Finally, Figure 16 and Figure 17 are the Spearman versions of Figure 7 and Figure 10, respectively. Overall, we find that Figure 16 and Figure 17 are consistent with their corresponding Pearson versions.

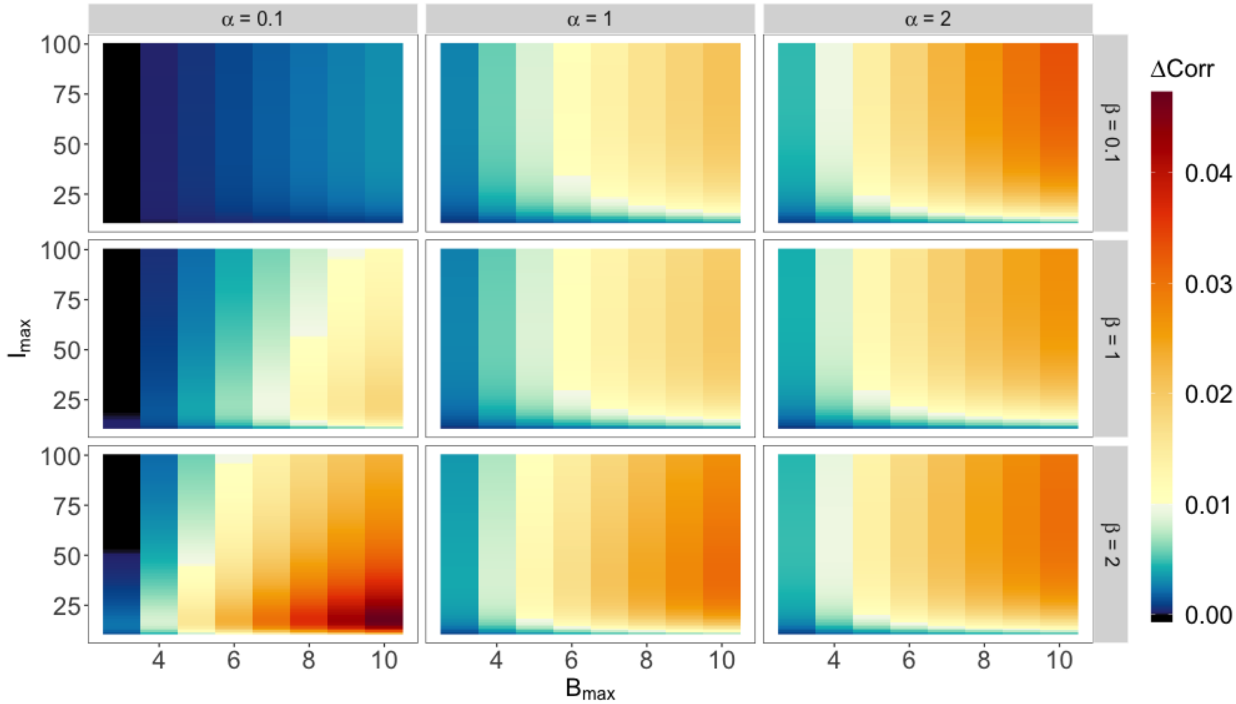

Figure 12 **The amounts by which fair skill adjustment improves the Pearson correlation between the experts' sample ($N = 50$) and long-term scores. The improvement is positive, except when α and B_{max} are both small, i.e., when the events are close to being certain, and there is very little heterogeneity in the number of potential outcomes of the events.**

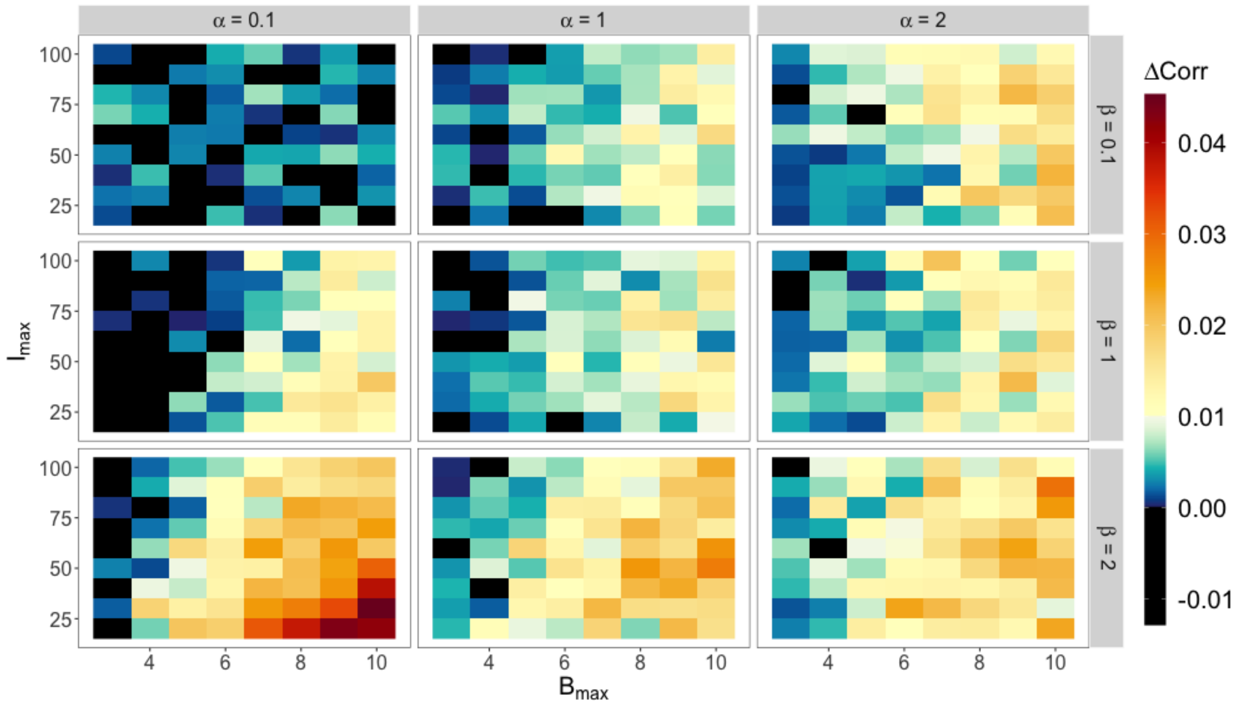

Figure 13 **The amounts by which fair skill adjustment improves the Spearman correlation between the experts' sample ($N = 10$) and long-term scores. The improvement is positive, except when α and $B_{\max}$ are both small, i.e., when the events are close to being certain, and there is very little heterogeneity in the number of potential outcomes of the events.**

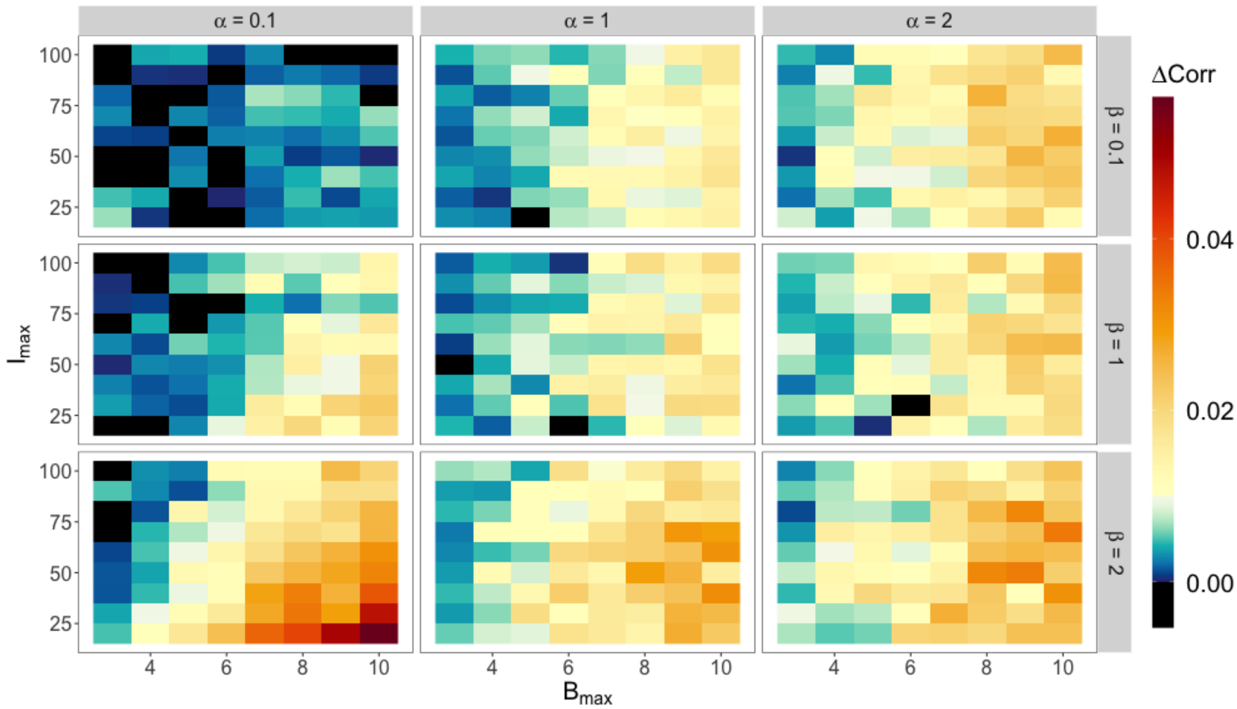

Figure 14 **The amounts by which fair skill adjustment improves the Spearman correlation between the experts' sample ($N = 20$) and long-term scores. The improvement is positive, except when α and B_{max} are both small, i.e., when the events are close to being certain, and there is very little heterogeneity in the number of potential outcomes of the events.**

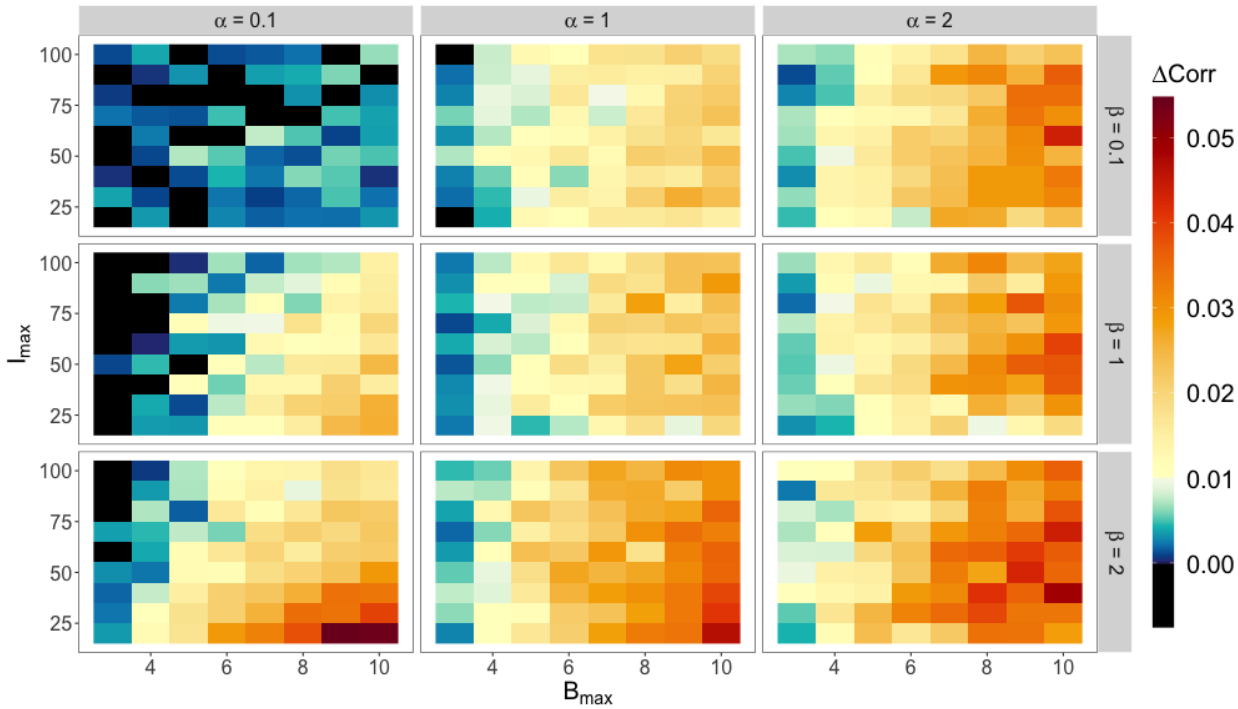

Figure 15 **The amounts by which fair skill adjustment improves the Spearman correlation between the experts' sample ($N = 50$) and long-term scores. The improvement is positive, except when α and B_{max} are both small, i.e., when the events are close to being certain, and there is very little heterogeneity in the number of potential outcomes of the events.**

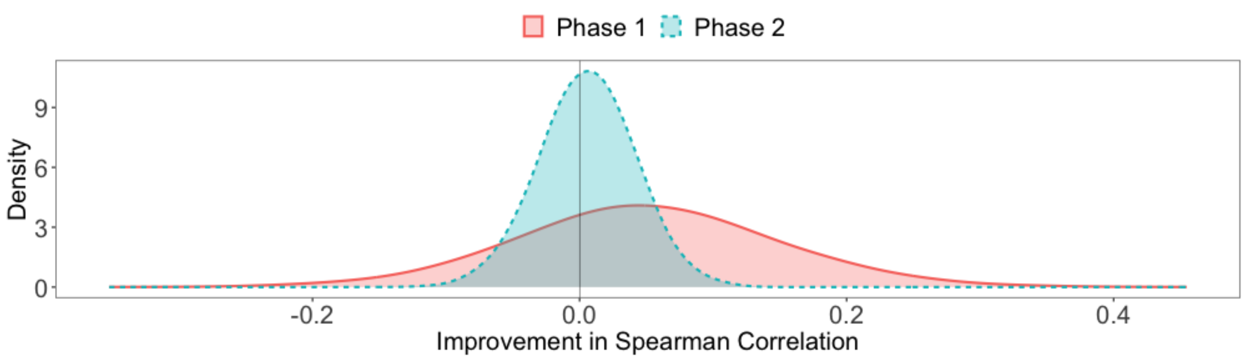

Figure 16 **The improvement in the Spearman correlation coefficient between the experts' sample and long-term scores by the fair skill adjustment.**

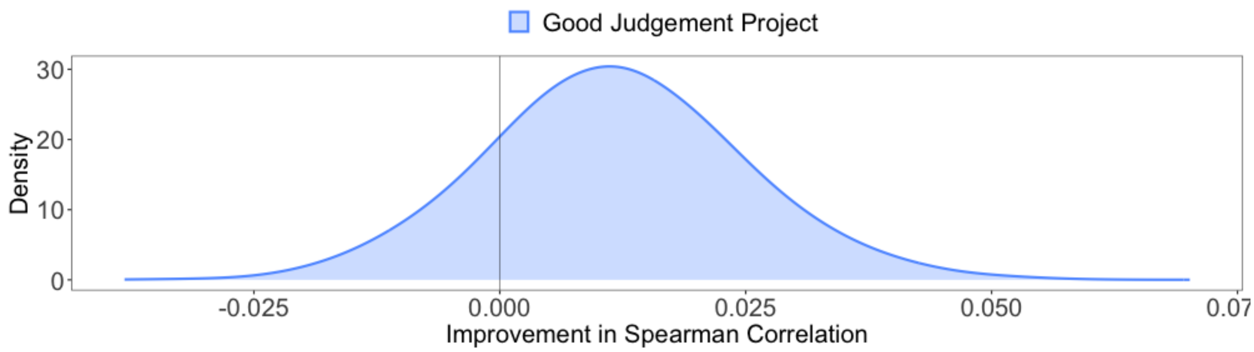

Figure 17 **The improvement in the Spearman correlation coefficient between the experts' sample and long-term scores by the fair skill adjustment.**

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